ADMINISTRATORS' PERCEPTIONS OF THE IMPLEMENTATION OF A DATA DASHBOARD IN ONE TEXAS CHARTER SCHOOL SYSTEM

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by

Erdal Caglar

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ADMINISTRATORS' PERCEPTIONS OF THE IMPLEMENTATION OF A DATA DASHBOARD IN ONE TEXAS CHARTER SCHOOL SYSTEM

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ABSTRACT

ADMINISTRATORS' PERCEPTIONS OF THE IMPLEMENTATION OF A DATA DASHBOARD IN ONE TEXAS CHARTER SCHOOL SYSTEM

by

Erdal Caglar

This qualitative, phenomenological narrative research study was designed to explore the perceptions of administrators regarding the implementation of a data dashboard system in one Texas Charter School System. The following questions guided the research: What do school system administrators perceive as benefits to implementing a data dashboard system in terms of administrators, teachers, parents, and students? What do school system administrators perceive as ways that a data dashboard system supports data-informed decision-making? What topics should the dashboard system encompass that are currently not being addressed? Open-ended interviews were used to collect data concerning the experience of 10 purposefully selected administrators in one charter school system in Texas. Findings suggested the following:

- Administrators greatly benefited from the implementation of the data dashboard.
- The data dashboard provides an advantage to its users by providing easy access to all data in one place.
- This study affirms that administrators use the data dashboard to monitor the academic progress of all students, enrollment numbers, and attendance rates.
- The data dashboard helps administrators make better in decision-making.

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Chapter I

Introduction to the Study

Bremser and Wagner (2013) expressed that data is meaningless by itself. They stated that when data is displayed on a form with graphics and summary, it becomes information understandable by everyone. According to West (2012), the dashboard provides all stakeholders with the critical information they need to effectively interpret and act on data in order to keep everyone on track to student, school, and district-wide goals. He argued that, for this purpose, when data is put into a form that is easily understandable, it becomes information. When information is used to guide decisions that are in the best interest of the students and families we serve, it becomes knowledge (West, 2012). Web-based business tools can help school leaders make sense of critical information and get a better handle on the key measures that drive student achievement (Carr, 2008).

Data-driven decision-making has become an essential component of educational practice across all levels, from chief state school officers to classroom teachers, and has received unprecedented attention in terms of policy and financial support (Mandinach, 2012). Moreover, Kowalski, Lasley, and Mahoney (2008) explained that school principals have great responsibilities on data-driven making processes at their schools. On the other hand, Brynes (2012) stated that most of the executive leaders become lost within a lot of information and are seeking ways to simplify the reporting tools in an understandable and summarized format. One of the most difficult of an executive's tasks is managing the strategic decision making (Ballou, Heitger, & Donnell, 2010). To meet this need, many companies have developed executive-level dashboards (Brynes, 2012).

Background

Kowalski, Lasley, and Mahoney (2008) stated that school principals have great responsibilities on data-driven making processes at their schools. Hallinger (2003) also recommended that school officials ask themselves what structures and programs they prefer and need for their students to become responsible and successful citizens of this information-age society. Blau and Presser (2013) stated that online school information management systems (SIMS) have become an important tool for effective management of pedagogical information and student data. They defined e-Leadership as the capability of a leader to engage others so that they can work collaboratively in technology-based classrooms. According to Blau and Presser (2013), in order to enhance e-Leadership, school leaders should follow two important steps: (a) the use of SIMS among students, teachers, school leaders, and parents, and (b) assign e-Leadership tasks to school personnel and actively watching its use at school.

Bremser and Wagner (2013) expressed the view that since dashboards provide a quick and easy access to real-time information to its users in order to interpret the data effectively, the use of customized user-friendly dashboards has recently increased. Moreover, according to Few (2006), dashboards have become popular in recent years as uniquely powerful tools for communicating important information at a glance. Ballou et al. (2010) also explained that data dashboards interactively provide key performance indicators and any other relevant information to executives in real time and in an easy format.

Tomic and Milic (2013) explained that when data is not well presented, it has no meaning. They suggested that when data is presented in a meaningful format it can

provide actionable insight (knowledge) into how business is doing. According to Bremser and Wagner (2013), creating a dashboard involves defining the dashboard objective, defining the dashboard metrics, seeking user input, building initial dashboard and testing, publishing the dashboard, and monitoring its use. They also emphasized the importance of dashboard visual design and gave guidelines for this purpose. Bremser and Wagner (2013) recommended that an effective dashboard have several important key elements and measures in order to provide focus for its users. On the other hand, they stated that the use of too many performance indicators may leave the dashboard users overwhelmed and have them dealing with challenges to process the data for an effective data-driven decision. They also indicated that when working on processing data, most users have limitations in terms of focus and memory. Because of this, having a dashboard with effective reporting capability can provide access to its users on additional performance indicators, or the data behind these indicators, to its users (Bremser & Wagner, 2013).

According to Lange, Range, and Welsh (2012), school leaders and teachers need all different types of data in order to interpret and act on data about students' progresses such as student achievement data, demographic data, school process data, and perception data. Following this further, Bernhardt (2007) and Reeves (2005) showed that it is very important for school leaders to facilitate professional development for their teachers in terms of data-driven decision-making processes. Moreover, they recommended school leaders to ensure professional development urges forward instructional change in the purpose of effective data-driven decision-making. In addition, Lange et al. (2012) argued that school leaders must consider individuals who are used to earlier processes so they can successfully develop a school culture that welcomes the effective use of data.

Problem Statement

School leaders and teachers should use a wide variety of data to form conclusions about student learning such as student learning data, demographic data, school process data, and perception data (Lange et al., 2012). Shen (2012) stated that by offering principals a lens for viewing their work related to monitoring and assessing schools' factors associated with student achievement, the instrument reinforced the importance of the principal's monitoring role and the importance of using a broad range of data sources to carry out that role.

Creating a data dashboard will help school leaders in this purpose (Ballou et al., 2010). He argued that school leaders need a data system that provides real-time information to users, including students, through customized user-friendly dashboards. When this information is used to guide decisions that are in the best interest of the students and families we serve, it becomes knowledge (West, 2012). This study is aimed at understanding the perceptions of administrators about the implementation of a data dashboard and to give other charter school administrators guidance on how to best use a data dashboard to improve the quality of education for all students in their school settings.

Statement of the Purpose and Research Questions

The purpose of this qualitative phenomenological narrative study was to explore the perceptions of administrators regarding the implementation of a data dashboard system in one Texas Charter School System. For the purpose of this study, a data dashboard system is defined as a tool to share key performance indicators and any other

relevant information with all stakeholders in real time and in an easy format (Ballou et al., 2010).

The following research questions will be used in this study:

1. What do school system administrators perceive as benefits to implementing a data dashboard system for administrators, teachers, parents, and students?
2. What do school system administrators perceive as ways that a data dashboard system supports data-informed decision-making?
3. What topics should the dashboard system encompass that are currently not being addressed?

Rationale of the Study

The findings from this study provided information regarding how charter school administrators perceive the implementation of a data dashboard system in one Texas Charter School System. Understanding these perceptions about this implementation will guide other charter school administrators about how to best use on using a data dashboard to improve the quality of education for all students in their school settings. According to West (2012), by itself, data have no value. He also reported that when data are put into an easily understandable form, data become information. When information is used to guide decisions that are in the best interest of the students and families that educators serve, it becomes knowledge (West, 2012). Ballou et al. (2010) argued that many companies need a data dashboard to identify and measure risk factors to help them make decisions. Bremser and Wagner (2013) explained that dashboards have become a popular tool for giving users better access to crucial information in a way that does not overwhelm them.

According to Ballou et al. (2010), school leaders need a system that provides real-time information to users, including students, through customized user-friendly dashboards. Creating a data dashboard could potentially help school leaders in this purpose (Ballou et al., 2010).

Bremser and Wagner (2013) expressed that isolated data are meaningless. Therefore, when data are displayed on a form with graphics and summary, data become information understandable by everyone. According to the researchers, the dashboard provides all stakeholders with the critical information they need to effectively interpret and act on data in order to keep educators on track to student, school, and district-wide goals. To support this, West (2012) contended that when data are put into an easily understandable form, data become information. When information is used to guide decisions that are in the best interest of the students and families served, data become knowledge. Web-based business tools can help school leaders make sense of critical information and get a better handle on the key measures that drive student achievement (Carr, 2008).

Assumptions

Pyrczak and Bruce (2005) clarified that assumptions are those items the researcher takes for granted as being true relative to the study. In relation to the present study, there are several assumptions that were made for this study. The following assumptions were associated with this study:

- All study participants answered the interview questions openly and honestly.
- Participants were able to read and clearly understand the interview questions.

- The research questions were adequately developed to understand the perceptions of the participants regarding the implementation of a data dashboard system.

Limitations

Creswell (2013) defined limitations as "potential weaknesses or problems with the study identified by the researcher" (p. 199). This study has the following limitations:

- The researcher chose to focus only on one charter school system having about 46 principals, 92 assistant principals, and six area superintendents in Texas. Only 10 of them were interviewed. Public school administrators were not included because the focus is one Texas Charter School System.
- Due to the limited nature of the study, it may be difficult to generalize the findings.
- The study was limited by the beliefs of the principals interviewed and their willingness to express those beliefs.
- The researcher is one of the executive members of this charter school setting, which may have limited the responses of participants, due to the professional relationship.

Delimitations

Delimitations were factors employed by the researcher to narrow the focus (Creswell, 2013) and specify the boundaries of the research project (Simon & Goes, 2010). This study encompasses the following delimitations:

- The study was limited to a purposeful sampling of charter school principals, assistant principals, and area superintendents.

- The study included participants with at least two years of experience in a charter school setting.
- This study included one charter school district.

Definitions of Terms

The relevant terms used in this research study were defined as follows:

- *Dashboard:* Dashboards display key performance metrics in a consolidated format (Ballou et al., 2010).
- *Data mining:* Data mining is the computer-assisted process of digging through and analyzing enormous sets of data and then extracting the meaning of the data (Hsieh, 2013).
- *e-Leadership:* e-Leadership is the ability of a person to influence the behavior of others in a digital technology-mediated environment (Blau & Presser, 2013).

Summary and Organization of the Study

In this chapter, the researcher introduced qualitative phenomenological narrative study about exploring the perceptions of administrators regarding the implementation of a data dashboard system in one Texas Charter School System. The background of the problem was presented along with the problem statement and purpose of the study. Three research questions were presented along with the rationale for the study. Finally, assumptions, limitations, delimitations, and definitions were presented. Chapter II contains the literature review covering what is presented about a data dashboard in the literature. The methodology was discussed in Chapter III. Epoche was included in Chapter IV. Participants' individual stories are presented in Chapters V through XIV.

Chapter XV includes a narrative analysis. The final chapter, Chapter XVI, discusses the Summary, Conclusions, Implications, and Recommendations for future study.

Chapter II

Review of the Literature

The purpose of this qualitative phenomenological narrative study was to explore the perceptions of administrators regarding the implementation of a data dashboard system in one Texas Charter School System. For the purpose of this study, a data dashboard system was defined as a tool to share key performance indicators and any other relevant information to all stakeholders in real time and in an easy format (Ballou et al., 2010). The literature review includes a number of key elements of a data dashboard. These elements include the data dashboard, creating the data dashboard, filling the dashboard's population with data, and determining a foundation of associations between the dashboard items, predicting, and situations. It will also include an association to monetary values, multiple dashboards' objectives, vital functional characteristics of dashboard, effective data use, and data informed decision-making. The literature review will also discuss early warning systems, course achievement, attendance, academic abilities, methods in establishing early warning data systems, data mining, outside the schoolroom data mining, e-Leadership, and technology to support data collection. It also aims to synthesize each of these components to determine the effectiveness of data dashboard usage for school leaders.

Data Dashboards

The results of multiple studies indicated that proper counseling, advocacy, and time are vital elements for instructors to positively utilize data as an instrument to guide schoolroom directives and increase student achievement (Lachat & Smith, 2005; Wayman, 2005). According to Wayman (2005), dashboards had a digital appearance and

were used by district instructional advocacy groups to engage in actions with instructors. The advocacy groups and instructors collaborated effectively and established some revisions on the initial appearance of a data dashboard (Lachat & Smith, 2005). Debush, Brown, and Killough (2003) suggested that the dashboard is an important tool for effective instructional advocacy.

The first process used in implementing a data dashboard includes developing an environment that is used as an instructional judgment-establishing instrument (Datnow, Park, & Kennedy, 2008). Multiple researchers proposed that this progression relied on the advocacy and anticipation that the school and district leadership established and implemented a data-oriented vision (Datnow et al., 2008; Wayman, Stringfield, & Yakimowski, 2004). Only limited research in technical journals regarding the interaction between motivation and digital dashboard use exists (Pauwels, Ambler, Bruce, Lapointe, Reibstein, & Skiera, 2009; Wind, 2005). In addition, application phases or application (Muntean, Bologa, Bologa, & Florea, 2011; Pauwels et al., 2009) choice of matrix (Debush et al., 2003; Pauwels et al., 2009) and their design (Yigitbasioglu & Velcu, 2012) all present difficulties for study because of the dearth of information available. This is especially true for dashboard expertise, which is the basis for this particular vein of research. Moreover, it was revealed that user output and dashboard applications were influential instruments in the management of performance in objectives such as decision-making justification and communication (Datnow et al., 2008). Wind (2005) noted that the worth of the data in dashboards did not show an alarming situation. As anticipated, it appeared a vital component for its application.

Bremser and Wagner (2013) expressed that because dashboards provide quick and easy access to real-time information for its users, the usage of customized, user-friendly dashboards has recently increased. Moreover, according to Few (2006), dashboards have become popular since their inception because of their ability to collect and present pertinent information to the user with minimal complexity. Ballou et al. (2010) also explained that data dashboards interactively provide key performance indicators and any other relevant information to executives in a format that is in real time and easy to use.

Creating data dashboards. Tomic and Milic (2013) explained that when data is not well presented, it has no meaning. They added that data only becomes meaningful when it can provide actionable insight about how business is doing. According to Bremser and Wagner (2013), creating a dashboard involves defining the dashboard objective, defining the dashboard metrics, seeking user input, building initial dashboard and testing, publishing the dashboard, and monitoring its use. Bremser and Wagner (2013), along with Byrnes (2012), also emphasized the importance of the dashboard's visual design and gave guidelines for this purpose.

Bremser and Wagner (2013) recommended that an effective dashboard have several important key elements and measures in order to provide focus for its users. On the other hand, they stated that the use of too many performance indicators might leave dashboard users overwhelmed, which will make processing data-driven decisions more challenging. According to the researchers, when working on processing data, most users have limitations in terms of focus and memory. For this reason, having a dashboard with effective reporting capability can provide access to its users on additional performance indicators, or the data behind these indicators.

Three extensive studies have been conducted on the creation of the data dashboard. Gulati and Oldroyd (2005), Reibstein, Joshi, Norton, and Farris (2005), and Wyner (2008) demonstrated collectively that the creation of a data dashboard is compromised of five separate stages. The following stages have been used to classify data dashboards: choosing the fundamental metrics, dashboard's population with data, foundation of association between dashboard items, predicting and substitutions, and association to monetary values (Reibstein et al., 2005).

Choosing the fundamental metrics. In the first stage of the creation of a data dashboard, Tim (2003) noted that there are two fundamental methods to metrics choice: tailored and general approaches. The general method contains the quantity of metrics that lower the limits on the virtual setup; this allows the data dashboard to maintain benefits of comparability, while conceding benchmarking through corporate segments, organizations, sectors, and time scales. Metrics, as revealed by Tim (2003), involved three profit and loss factors about profit, revenue, and marketing expenditures. The three factors involved were the number of clients, aligned cash flow, and retention rates that appeared in the head of revenue (Tim, 2003).

In their appraisal of associations and service domains of study, there were numerous significant metrics suggested by Rust and Chung (2006). According to Rust and Chung, client oriented metrics include client anticipations, fulfillment, issues, as well as worker oriented metrics including worker fulfillment and output. Thus, those need to be the segment of a dashboard for service entities (Clark, Abela, & Ambler, 2006; Rust & Chung, 2006).

In comparison, the tailored method claims that each corporate unit or organization keeps its personal plan, positioning, and demands as distinguished metrics to locate output for its particular goals (Rust & Chung, 2006). Kamakura, Mittal, Rosa, and Mazzon (2002) noted that this method welcomes a greater level communication through the workplace, while explaining significant elements about the business and identifying the factors considered extremely important for numerous organizations. According to Paine (2004), individuals who used these systems are uniquely qualified to help design them; these people know what they are interested in within the system and what capabilities the system should have to be effective and efficient (MarketingNPV, 2005).

This method is best with the creation of data dashboard advantages of cross-functional teamwork and investigation (Ingram, Louis, & Schroeder, 2004). Nevertheless, the tailored method required much more attention and used substantial time that entities get stuck at this segment, or create numerous metrics, with minor anticipation of decreasing them, as a department or professional employee prefers each set (MarketingNPV, 2005). Engaging too many departments in the executive administration of these metrics effectively blocks mandatory generalization within the dashboard program which reduces the application's ability to fulfill its purpose within the educational environment (Clark et al., 2006). Maintaining metrics at a controllable quantity demands executive and administrative leadership and requires exploring the important factors of performance (Pauwels & Joshi, 2008).

Dashboard's population with data. In the second stage of the creation of a data dashboard, the application of recently existing metrics engages the procedure at an insignificant monetary value, generally through investigating favorable substitutions and

uniting current factors (Lehmann & Reibstein, 2006). According to Lehmann and Reibstein (2006), waiting for influential and important metrics can be a significant in the long-term. For example, in the early 2000s, a United Kingdom financial institution, Abbey National, switched from utilizing monetary computation of the brand considering client-oriented factors on the grounds of cognitive chasing (McIntire, 2002). As the monetary factor appeared computable, the administration determined that it must be faulty (Pauwels & Joshi, 2008). Furthermore, they determined that the client-oriented factors developed extra actionable awareness. Similarly, Unisys investigated its dashboard application and altered advertising from reportage solely on competence metrics (Tim, 2003). Unisys looked at the production of leaflets and compared that number to the previous period with a similar number of workers (MarketingNPV, 2005). This comparison was designed to investigate corporate efficiency factors (MarketingNPV, 2004; Miller & Cioffi, 2004).

The population of the dashboard with data is nevertheless unimportant. Service organizations gather large amounts of data from multiple foundations; possibly the greatest issue at play is the periodicity of the data (Debush et al., 2003). Little of the data can be gathered from deals reports, while additional data points are gathered yearly, or even less frequently (Debush et al., 2003). For example, brand equity factors are gathered very infrequently (Nelson, Todd, & Wixom, 2005). Debush et al. (2003) suggested that few metrics showed significance compared to others, therefore notices need to be considered to be urgent; this is a fundamental function of the data presentation of the dashboard.

Foundation of associations between the dashboard items. The third stage incorporates defining the fundamental associations among the metrics (Wyner, 2008). This stage changes the dashboard from a convenient demonstration of information to a more profound comprehension of the trade and decision support system (Rust, Lemon, & Zeithaml, 2004). Metrics not only meet cause-and-effect associations for strong attribution of demonstration amendment or the abilities to produce and check substitution mix provisions; they also allow for forecasting and determining future movement (Spillane, Halverson, & Diamond, 2001). Rust et al. (2004) developed a process that suggested that challenging marketing plan choices to be transacted off on the grounds of forecasted monetary profit. This monetary profit is operationalized as the alteration in an organization's client fairness metric comparative to the improving expenses mandatory to generate the alteration (Rust et al., 2004). Thus, these researchers discovered associations among the client equity factors including occurrence of group purchases, average number of purchases, brand switching designs, and organization's contribution margin through predicting a brand switching matrix utilizing a logistics choice model (Kamakura et al., 2002).

Additional techniques to determine matrix associations involve structural equation models and connection assessment (Kamakura et al., 2002; Schneider, Ehrhart, Mayer, Saltz, & Niles-Jolly, 2005). The associations between advertising attempts, dashboard metrics, and presentation were possibly categorized by indirect influences, direct causativeness, and performance response (Dekimpe & Hanssens, 1999). It is therefore important that a diverse system method should reflect a logical, albeit complex, opportunity for predicting these associations as utilized through studies conducted by a

number of researchers (Pauwels & Joshi, 2008). Otherwise, administrative divisions through decision calculus may be utilized to predict numerous of the associations (Little, 1970). In their landmark research on the calculus of associations, Little (1970) suggested that the final outcome is to elucidate and assist the balancing of the multiple suppositions under which administrators are engaged in operations.

Upon the availability of data, it is expected to check the suppositions that contribute in carrying out corporate judgments (Danaher, 2002). Extensive doubt relevant to the decisions or difference amid the administrative group can reveal benefits to begin study, and with respect to data gathering, or investigation, to effectively comprehend the associations (Cromney, 2000). To provide a real-world example of this data presentation, consider an organization engaged in telecommunication services that establishes market investigations to investigate the impact of multiple pricing plans (Danaher, 2002). Vendors of credit cards may utilize investigations to confirm the impacts of multiple loyalty reward schemes (Wirtz, Mattila, & Lwin, 2007). Danaher (2002) also suggested that because the dashboard encourages the clarity and reiteration of subject investigations, it is convenient to facilitate administration devotion to then involve them in comprehensive size-marketing schemes. When these individual workers are moved about within the company, they will leave their dashboard active for the next individual in their position instead of sending off the liability of establishing a comprehensive system or mental model from the beginning (Danaher, 2002). According to Kaplan and Norton (2000), a strategy map was helpful for this particular situation; it provided the participating researchers with the information they needed in a very accessible, efficient manner.

Predicting and situations. The fourth stage used the dashboard's fundamental model to scenario development and financial plan establishment (Wierenga & Bruggen, 2000; Wind, 2005). If the organization enhances the precision with which it predicts service requirements through the utilization of data related to previous sales forms, it needs to improve profits as stated by (Radas & Shugan, 1998). Moreover, utilizing what-if analysis delivers administrators with an instrument for assessing advertising output (Rust et al., 2004). Zeithaml, Bolton, Deighton, Keiningham, Lemon, and Petersen (2006) proposed that this declaration is not acknowledged in numerous recent dashboards which emphasize particularly on reportage of recent operations.

Association to monetary values. The fifth stage associated advertising expenses entirely through the intermediate advertising metrics against deals and to the monetary consequences for the organization (Lehmann & Reibstein, 2006; McGovern, Court, Quelch, & Crawford, 2004). According to Srinivasan and Hanssens (2009), brand equity, client fulfillment, client equity, and advertising actions including the organization's marketing were all associated to the worth of an organization. For example, multiple studies analyze the impact of innovative product declarations and overviews, client fulfillment, marketing, and incorporating innovative networks (Fornell, Mithas, Morgeson, & Krishnan, 2006; Pauwels, Jorge, Srinivasan, & Hanssens, 2004; Pauwels & Joshi, 2008).

The above five stages used in creating data dashboards procedures are not the sole tracking method toward the perfect stage where administrative goals, plans, and corporate models are comprehensively expressed in an operational dashboard (Srinivasan & Hanssens, 2009). There are substitutes for these five stages available in the literature as

well, particularly in the studies conducted by Wyner (2008). The fundamental change shows that in the fifth stage the emphasis is put on incorporating chartered accountants, while clients receive promotions from researchers (Wyner, 2008). As lines are the vital developers of an entity's cash stream, the client and stakeholder worth views are naturally associated (Wyner, 2008; Pauwels et al., 2004).

Technology to support data collection. Kowalski et al. (2008) suggested that the gathering of data, which includes the process of data gathering and authorizing individuals' access to the data, are significant variables for the positive application of data. While noticing the application of technology, Garrett (2013) declared that school administrators need to be focused on the goal of effective behavior. Furthermore, Garrett (2013) emphasized that confidential data storing and safety of classifying knowledge in e-forms is an additional notice for school employees.

The confidential details related to students that schools retain are safeguarded by The Children's Online Privacy Protection Act of 1998 (COPPA) and The Family Educational Rights and Privacy Act of 1974 (FERPA) (Center on Education Policy, 2004). The Public Forum on Education Statistics 2006 reveals an impression to school employees related to the safety of academic data (Center on Education Policy, 2004). The organization declared that the group of words such as academic records is outlined as record, document, file, and additional stuff possessing details associated to a scholar or student; these records can be managed through the academic entity, or via an individual performing for such entity according to 34 CFR § 99.3 (Macfadyen & Dawson, 2010). It involves each detail of an individual's scholastic career, irrespective of medium, incorporating information stored on audio or videotapes, in script, on computer or

electronic records, in design, on microfilm and microfiche, and on film (Center on Education Policy, 2004). Additionally, according to researchers like Marsh, Kerr, Ikemoto, Darilek, Suttorp, and Zimmer (2005), who studied in a national school region, each piece of data related to services are procured under the Individuals with Disabilities Education Act (IDEA).

According to researchers Arnott and Pervan (2005), dashboards refer to a specific kind of decision support system, and may be outlined as a shared performance management and visual instrument. This instrument can provide a plethora of information that an organization can utilize to recognize, explore, and communicate issues that require action (Yigitbasioglu & Velcu, 2012). Dashboards involve practical and visual characteristics that assist the user in enhancing explanation and reasoning (Mason, 2002). According to Yigitbasioglu and Velcu (2012), the features can be used by multiple operators. Frontline employees track inventory, middle executives assess lagging measures, and executive administrators analyze performance against long-term goals.

According to Avidov-Unger and Eshet-Alkalai (2011), multiple variables contributed to meaningful growth in dashboard and dashboard technologies. Initially, the developments in information technology uncovered the technical infrastructure to establish them, which fared positively in the corporate domain. Also, the willingness of individuals to implement the new technology contributed to growth. Kaplan and Norton (2005) added that programs were an innovative method of gauging performance with the outline of data they produced. Lapointe (2005) stressed that variables such as requiring cross-department incorporation and administrative prejudice determined dashboard acceptance and use.

In recent years, the focus of dashboards has moved from checking performance to more analytical goals, including situation assessment, drill down competences, and presentation format suppleness (Yigitbasioglu & Velcu, 2012). According to Yigitbasioglu and Velcu (2012), format suppleness is the inclusion of multiple types of data presentation elements (like charts and tables). Card, Mackinlay, and Schneiderman (1999) clarified that with the fast advancement in corporate technology, dashboards will continue to innovate growth in the next period.

Chirichello (2010) listed the following theories to highlight the importance of dashboards through multiple categories of discipline: bookkeeping, information systems, and reasoning psychology. The implementation of psychology is often overlooked and misunderstood. Dashboards alleviate the executive's difficulties in the decision-making process via a procedure known as visualization, which is called shared visual illustrations of abstract, non-physically oriented records to intensify thought (Card et al., 1999). According to Vessey and Galetta (1991), this theory would promote the relationship between the judgment capabilities of an individual, the information performance format, and the job at hand. This relationship would deliver favorable considerations with reference to the selection of a performance format to be functional. For example, charts are quite favorable for latitudinal jobs that incorporate predicting and associating. They are also excellent for jobs that demand multidimensional data assessment and design acknowledgement (Card et al., 1999). Tabular knowledge is very influential for symbolic jobs and for operators who are inclined towards patience, such as financial forecasters and bookkeepers (Dexter, 2007).

The theory of cognitive fit shows that judgments established under the fit situation are premier associated to the non-fit (Vessey & Galetta 1991). For performance management jobs that include associations between numerous factors, known as *Key Performance Indicator,* and graphic representations provide strong explanations (Yigitbasioglu & Velcu, 2012).

Likewise, the fit between single skills, task provisions, and the functionality of the technology is promoted by task-technology fit theory (Goodhue & Thompson, 1995). For example, these are tasks that require knowledge from multiple units in the workplace, and they must all be incorporated in the dashboards of the workplace (Mehra, Smith, Dixon, & Robertson, 2006). In these situations, the dashboards must be designed according to the specific needs and functionality required for the tasks and operators (Nelson et al., 2005).

Multiple dashboards' objectives. A study conducted by Pauwels et al. (2009) proposed that the most used objectives for dashboards included checking, uniformity, scheduling, and communication. Checking was comprised of daily assessment of metrics that are expected to reveal a remedial action. According to Pauwels et al. (2009), checking may be considered the dashboard's greatest purpose. Boudett, Murnane, City, and Moody (2005) defined uniformity that assessment processes utilized across corporate units and departments are the same. Dashboards have also been utilized for scheduling, but are subject to the existence of scenario assessment among its characteristics (Blau & Hameiri, 2010). The dashboard reveals communicated values and performance of an entity to its stakeholders with the assistance of the selection of metrics (Pauwels et al.,

2009). Only some of the outlined objectives are utilized by professional entities (Clark, Abela, & Ambler, 2006).

Doll and Torkzadeh (1998) discovered that dashboards could be applied in the Management Information Systems (MIS) field to aid workers in tasks such as incorporation with problem-solving and client service. According to Pauwels et al. (2009), administrators can use judgment and judgment justifying to determine whether data is appropriate for them or their executives. Self-monitoring, communication, and uniformity are resolution functions that are associated with the dashboard's job completion and progress feature (Sutherland, 2004). According to Pauwels et al. (2009), when dashboards are incorporated in enterprise systems, operators can share information vertically with others.

Vital functional characteristics of dashboard. In recent years, software suppliers including Business Objects, Microsoft, Cognos, JasperSoft, and Qlik View competed in growing needs for dashboard demand (Neely, Yaghi, & Youell, 2008). Wierenga and Bruggen (2000) explained that although suppliers stress that the incorporation of dashboards can improve the performance of an organization in multiple aspects such as customer satisfaction, profits on investments, and positive cash flow, organizations look for easy substitutes. Therefore, dashboard critics will use Excel spreadsheets instead (Neely et al., 2008).

There are a number of characteristics that make a dashboard functional from the point of view of the researchers (Wierenga & Bruggen, 2000). The following are functional characteristics of dashboards: real-time notices and alerts, drill down competences, situation examination, presentation suppleness/theory directed format

choices, and exterior benchmarking (Pauwels et al., 2009; Ying, Lijun, & Wei, 2009; Yigitbasioglu & Velcu, 2012). Real-time notices and alerts were mandatory for taking remedial action when there was a deviation from the established corporate goals (Ying et al., 2009). Alerts may be executed through the application of different colors, blinking signals, or auditory signs (Pauwels et al., 2009). According to Ying et al. (2009), the trickle-down notification is important because it notifies operators that a comprehensive assessment needs to be conducted. The drill down feature is highly desirable as it allows users to slice and dice data for a more detailed analysis without which the users would be forced to switch to another reporting tool (Bariff & Lusk, 1977; Benbasat & Dexter, 1979). Situation examination is a favorable tool, particularly for corporations who use dashboards for purpose of planning (Benbasat & Dexter, 1979). Operators use this tool like a judgment-advocating instrument to observe how changes in particular factors influence other factors. Operators may also use it for emphasizing and revealing the significance of dominant factors (Zeithaml et al., 2006).

Theory-directed format choices are associated to the prior item as presentation suppleness to the level that the system confirms on the greatest favorable data setup by taking notice of the task and operator background despite the conceding the opportunity to the operator (Cohen, 2003). Ultimately, exterior benchmarking may allow operators to see the performance of their organization and compare it to its rivals (Huber, 1983). There are three fundamental elements found in the above characteristics (Cooley, Shen, Miller, Winograd, Rainey, Yuan, & Ryan, 2006). These elements include: drill down, situation examination, and presentation suppleness elements. It is expected that these

variables will contribute to dashboards becoming a diverse and influential instrument, favorable for multiple operators and situations (Pauwels et al., 2009).

Data Mining

Data mining is the process by which computers are used to analyze large sets of data and determine patterns, correlations, or causations within those data sets (Hsieh, 2013). Goren (2012) postulated that data mean different things to different people in different settings. Moreover, Spillane (2012) noted that the manner in which data and evidence is delivered to people in school influences what aspects of data practitioners notice and attend to.

Creswell (2013) explained that data specialists do not solely gather their personal data while consciously explaining exteriors' data and the culminations, but also approach saved data to apply it in enhancing their repetition (Creswell, 2013). Data mining in education helps researchers examine raw data in order to understand the learning and teaching procedure (Colby, Smith, & Shelton, 2005). Numerous methods of examining particular kinds of raw data is present in the field of education, each reflecting possible application for instructors, analysts, administrators, and students themselves (Creswell, 2013; Colby et al., 2005).

Creswell (2013) presented a number of questions regarding an instance of data mining. He points to assessment that shows variables linked with meaningful advantages or disadvantages in consistent achievement trials: Do greater deviations initially show notices in the third rank? Do greater benefits associate to a particular prospectus? Does presence associate with achievement? Does presence associate to achievement deficiencies among students? Does any factor change this association? Evers and

Walberg (2002) noted that the answers to these questions can be determined via data mining. Though the fundamental objective of data mining is to enhance educational exercise and student knowledge, not check theory, it is probable that forms in the data are expected to appear to breed innovative theories or update the previous ones (Evers & Walberg, 2002).

Outside the schoolroom data mining. Data mining is significant as it concedes data to operate outside the classroom with regular utilizations in entire schools, states, districts, and regions (Sawchuk, 2010). A study conducted by Sawchuk (2010) took notice that the group of words "from assessment to action" revealed the behavior of a single high school based on Fall River, Massachusetts (Sawchuk, 2010). This high school achieved assessment to action through investigating indicators of high school students' disconnection such as goal factors of absence, failure, retention, and dropout and associating favorable tendencies in those records with alterations appearing from an assessment of straight communications with guardians, instructors, and the students themselves (Reeves, 2005). Issues with directives in Advanced Planning (AP) categories appeared obvious through assessment of transcripts reflecting A and B ranks associated to dropouts on the AP examinations, and this result showed a precise appeal to act for the advanced school and revealed findings for changing the inflation in rank with educational consistency (Dexter, 2007; Reeves, 2005; Sawchuk, 2010).

Whiting (2006) conducted research that illustrated entire district data mining in Lafourche, Louisiana. This research was done for the purpose of comprehending the reasons and outputs of students' discipline issues and school efficiency in safeguarding the students and their peers from additional unfavorable attitude. Indeed, 33,000

discipline findings involved a documented illustration of occurrences and school feedback regarding those incidents (Tan, 2010; Whiting, 2006). Although, this kind of qualitative record appeared difficult to assess, the school district was able to compile the information required to do data mining on the district's students (Whiting, 2006). Research conducted by Wayman (2005) suggested that a sole act of infraction and discipline can establish an unforgettable behavioral experience; nevertheless, it cannot reveal a long-term pattern of behavior for the student.

Cohen (2003) revealed that further instances of district factors including drilling, warehousing, and mining enhance education beginning with the supposition that experimental advocacy of the instructor is the significant potent contributor impacting student education. Cohen (2003) assessed numerous data foundations developed with the span of time. One of these data foundations was signifying data gathered over numerous periods for similar instructors and students. Another was longitudinal data with the objective of inventing dependable associations between quality of education and student learning, which means peer-to-peer education.

A sole instructor's matrix can involve marks on the state benchmarked papers, course ranks, subject paper marks, quantity of students, and students' performance indicators from former periods with each associated to entire school, entire-region, similar-schools, region, and national data (Cohen, 2003). Utilizing "cautious likeness" and potential instruments to rearrange the data, Cohen (2003) established particular instructors as significantly or insignificantly influential. Before making decisions on teacher effectiveness, the quantity of data needs to be analyzed (Cohen 2003; Cromney, 2000). The objective is to outline the best practices of highly effective teachers, then

share these practices with less effective teachers to assist in the goal of better student learning (Cohen, 2003). With the existence of multiple quantities of instructors, it is also probable that the instructor and the school administration as a whole will gain knowledge whether a particular educational instrument is linked with effective action among lower-achieving, average, or high-achieving students (Chirichello, 2010; Kowalski et al., 2008). The similar essence of investigation may be utilized to focus particular students at danger to drop out through estimated charts or trajectory with the objective of developing a plan that prevents this productivity (Datnow et al., 2008).

Nationwide data groups also utilize the strength of data mining. A study performed by Swanson (2010) illustrated losing odds in 21 urban regions in terms of graduation rates. The rates of graduation at the regional level may be estimated in a numerical process through information of particular variables like size, rural versus urban, expenditure, and socioeconomic and ethnic factors (Swanson, 2010). Through associating estimated with real graduation rates, the researcher is able to investigate regions that fail or beat their arithmetical estimation (Anderson & Dexter, 2005; Lachat & Smith, 2005). Moreover, research of these over or under behaving regions may uncover variables that differentiate them from the others (Spillane et al., 2001).

Instructors are also determining the importance of auxiliary sets of trial data, named as data exhaust by researchers (Creswell, 2013). According to Creswell (2013), data exhaust is a digitized information foundation that frames a trial mark and is gathered during management of the trial. Indeed, instances involve period spared on a sole item, amended keys, ignored queries, entire period spared, the location where feedbacks emerged haphazard, and additional same data foundations that express checking attitude

(Lachat & Smith, 2005). Assessment of data exhaust can demonstrate associations between data sets that are helpful in the explanation of a comprehensive trial mark (Creswell, 2013). The instruments and reserves for data mining do exist for numerous data specialists (Creswell, 2013). The software in advanced editions of MS Excel and XLMiner help school administrators treat the data (Creswell, 2013; Lachat & Smith, 2005; Kowalski et al., 2008).

Scholastic data require numerous methods such as quantitative, which includes numerical data sets, and qualitative, which encompasses video, text, and so on (Rathje & Murphy, 2001). A study conducted by Merriam (2009) uncovered numerous processes for data mining from physical records and reported that exclusive deviation in documented, digital, visual, and bodily factors (artifacts) exist for investigators for mining. Physical records are many and incorporate public documents, media booms, journals, letters, and visual products like photos. According to Rathje and Murphy (2001), physical records may also be mined, incorporating garbage data sets, containing the benefit of providing proof of real, unreported, attitude. School records share proof of, for example, of school or paternal incorporation. Online data foundations, (for instance, school websites or electronic mail) can also deliver information sources not conveniently accumulated elsewhere (Lehmann & Reibstein, 2006; MarketingNPV, Unisys Overcomes 6 Common Dashboard Mistakes, 2004; Reilly, 1982).

With the existence of widely available online data, it appears sensible that numerous qualitative studies are expected to mine text-rich foundations significantly associated to academic exercises and student features, like digital conversation clusters or programs, rule declarations, transcripts, and script samples (Rathje & Murphy, 2001).

Digital data access, gathering, and assessment, nevertheless, show contesting principled dilemma and preventions (Pauwels & Joshi, 2008).

Effective data use. According to Clark et al. (2006), data quality is known as the worth of the knowledge content shown on the screen of the dashboard. They also stated that data quality is a problem in dashboards, as per the examinations performed in 2001 and 2003 on established United States organizations. Studies conducted by Clark et al. (2006) outlined quality problems with reference to data established from application addition issues, which may be central to applying dashboard procured data and alternatives with further conventional instruments like MS Excel (Clark et al., 2006; Tomic & Milic, 2013;).

For the studies conducted in the domain of information systems, the outline of data quality is known as the format and content of data together developed by information systems (Gorla, Somers, & Wong, 2010). The studies conducted by Nelson et al. (2005) as well as Huh, Keller, Redman, and Watkins (1990) outlined quality by considering the expectations of data clients who are engaged in special tasks. The concepts of utilization of precision, wholeness, money, and uniformity for data quality were all applied by Huh et al. (1990). According to Huh et al. (1990), data is comprehensive on the basis of the provisions of users. Thus, for few operators, data is expected to address each expectation to address their tasks, while for additional operators, data may be incomprehensive (Huh et al., 1990). Currency is known as upgraded data and multiple operators possess requirements of upgraded information (Huh et al., 1990). There is a dependency of data currency on the operator perception and task. Huh et al. (1990) also stated that uniformity between data operators is very important. Otherwise,

the absence of disagreement between data operators effects the credibility of the data developed by the system (Huh et al., 1990).

In organizations, decision makers perform so many different tasks to arrive at numerous decisions (Reilly, 1982). Earlier studies conducted by Jacobson (2007) and Reilly (1982) revealed that under these different tasks, executives using these data sources came up with better effective decisions reliably, precisely, and quickly. One of the advantages of dashboard usage is the instant access to knowledge (Nelson et al., 2005; Pauwels et al., 2009).

According to Lange et al. (2012), school leaders and teachers need different types of data in order to interpret and act on data about students' progress such as student achievement data, demographic data, school process data, and perception data. Following this further, the Bernhardt's (2007) and Reeves' (2005) studies demonstrated that it is very important for school leaders to facilitate professional development for their teachers in terms of data-driven decision-making processes. Moreover, they recommended that school leaders ensure professional development promotes forward instructional change in the purpose of effective data-driven decision-making (Bernhardt, 2007). In addition, Lange et al. (2012) argued that school leaders must consider individuals who are used to earlier processes so that they can successfully develop a school culture that welcomes the effective use of data.

Data-driven Decision-making

After implementation of the No Child Left behind Act (NCLB), support for accountability and increased student achievement in the United States public schools did not improve (Slavin, 2002). Renowned academic experts strategically criticized

education as a domain in which school administrators establish judgments on the basis of perception, gut instinct, or fashions (Slavin, 2002). Educators of data informed decision-making actions claimed that influential utilization of data causes school systems to substantially gain information about themselves, determine contests and successes, locate domains of enhancement, and assist assessment of the efficiency of plans and practices (Mason, 2002). Indeed, the fundamental theory of NCLB demands that educators need to possess awareness to assess, explain, and utilize data in order to enable execution of informed decision-making in every domain of education, from individualized growth to student progress (Center on Education Policy, 2004; Slavin, 2002).

Earlier studies proposed that data informed decision-making contains the strength to improve student performance (Johnson, 1999; McIntire, 2002; Peterson, 2007). When classroom instructors are aware of data utilization, they are better able to appraise current instructional outcomes, investigate areas needing improvement, and effectively design strategies for enhancement (Earl & Katz, 2006). According to Center on Education Policy (2004), the latest available public research regarding the influence of NCLB investigated regions that are certainly engaged in assigning reserves to improve the application of student achievement data. These regions employed strategies such as a method to impart directives in schools investigated as requiring enhancements (Center on Education Policy, 2004; The Information Edge, 2006). Student achievement data may be utilized for numerous objectives, involving assessing growth toward regional and state benchmarks, observing performance and progress of students, confirming where reviews deviate and join, and examining the effectiveness of domestic prospectus and instructional practices (Crommey, 2000).

Nevertheless, data should be vigorously utilized to increase effective decisions in schools, and help schools facing ineffectiveness in terms of their capability to increase instructional quality (Ingram et al., 2004; Marsh et al., 2005). In order to meet this issue, regions financed in Management Information Systems and specialized growth strived to generate professionalism and capability at the level of the school (Borja, 2006).

Likewise, in the charter school field, academic service sources involving charter management organizations (CMOs) and education management organizations (EMOs) also attempt to establish effectiveness of school administrators on decision making in schools and through the network (Colby et al., 2005). Many of these institutions use data informed decision-making as one of their fundamental foundations. A case study of an Edison school established that the education management organization assisted in growing an environment of data application and data informed decision-making practice throughout its curriculum, evaluation, and organizational structure (Sutherland, 2004).

Regardless of systems oriented funds to establish capability and professionalism for data informed decision-making, numerous researchers determined that instructors are not engaged in progressively utilizing data to impart preparation and instructional judgments; instructors should be required not only to utilize data, but also to maintain the authorization and the tendency to do so (Earl & Katz, 2006). After the review of studies conducted in the domain of data informed decision-making, it is proposed that school level educators not solely require systematic advocacy, but also engage in substantial decision-making independence to establish site level judgments about data (Earl & Katz, 2006). Furthermore, Cohen (2003) established that professionalism and capability at the school site for data informed decision-making is mandatory but nevertheless not a

substantial situation for success. Ultimately, in establishing an accountability network, the disparity in the division of information between the schools and the central office needs to be addressed (Doll & Torkzadeh, 1998; Earl & Katz, 2006).

According to Evers and Walberg (2002), holding students, teachers, schools, and school districts accountable for student performance is a contemporary invention. With the accumulation of the vast amounts of data in education, much of the discussion about how to use data for decision making evolves from data-driven to data-based and to data-informed (Cooley et al., 2006; Knapp, Swinnerton, Copland, & Monpas-Huber, 2006; Shen & Cooley, 2008), with an understanding that data will inform rather than drive decision making because there are rational, political, and moral elements in decision-making, and data is only one of many important elements in the process. Shen and Cooley (2008) explained that by offering principals a lens for viewing their work related to monitoring and assessing schools' factors associated with student achievement, the instrument reinforced the importance of the principal's monitoring role and the importance of using a broad range of data sources to carry out that role.

Early warning system. Early warning data provides a method to meet an issue that outbreaks numerous failure deterrence exertions: positively focusing the right students (Pinkus, 2008). According to Macfadyen and Dawson (2010), pedagogically meaningful information can be extracted from Learning Management System (LMS) generated student-tracking data. They explained that these findings are helping to facilitate the development of a customizable dashboard-like reporting tool for school leaders that will display real-time data about student achievement and progress. This is particularly true for students who can be measured as being in danger of failure because

of demographic variables, but who are very effective in school and do not attain needless failure deterrence interferences (Pinkus, 2008). Alternatively, as Earl and Katz's (2006) studies suggested, with warning systems, students who are in danger of failing are likely to receive the assistance they need. Educators, activists, and policymakers can select to hold early warning data systems as instruments for the additional influential and efficient application of insignificant funding to enhance student productivity (Few, 2006).

For periods, researches revealed multiple variables that influence whether a student is at a higher or lower danger of failure (Sawchuck, 2010). In regions of extreme poverty situations, there were no systems that revealed grades, credits, and attendance as greatly predictive (Few, 2006; Gleason & Dynarski, 2002; Pinkus, 2008). On an extensive scale assessment of publicly financed dropout exertions, the researchers determined that dropout deterrence esteems regularly contribute to students who are not dropped out and do not contribute to students who dropped out (Gleason & Dynarski, 2002).

Course achievement. Research conducted by Pinkus (2008) and Sawchuck (2010) illustrated that course achievement can be the influential estimator of ultimate graduation. According to their research, students with a B average or higher in their freshman year have a 95% probability of graduating. Alternately, the research demonstrated that those freshmen that achieved a lower C average were inclined toward failure to graduate (Pinkus, 2008). This advocates the rational concept that deficiency of academic achievement is an obstacle to graduating from high school (Sawchuck, 2010). Luckily, numerous classroom plans exist that may utilize instructors to improve students' subject progression, procure additional grading periods, strike a management between

significance and thoroughness, and procure advocacy for students who are pursuing with capabilities or content (Sawchuck, 2010; Pinkus, 2008).

Attendance. Most researchers noted that one of the most important metrics for predicting student success is attendance (Pinkus, 2008). Pinkus (2008) reported that in an extensive longitudinal investigation, students who showed greater nonappearance, escape sessions in every seven days, or were absent ten or above periods in 30 days were six times more likely to fail than their peers. A recent study conducted in Chicago revealed that approximately 90% of freshmen who were absent fewer than seven days of school each semester progressed, regardless of previous year benchmark grades (Pinkus, 2008). Mioduser, Nachmias, Tubin, and Forkosh (2006) and Pinkus (2008) noted that students who fail also have significant attendance problems. They stated in their research that those who could not finish a school and earn a degree had missed numerous periods of school and faced difficulty addressing the academic issues that caused a failure.

Colby et al. (2005) noted that progressive high schools and their improvement models regularly welcome a long-term attempt that involves receptive and defensive plans together. A study that investigated plans for improving presence described them as dropping into four corresponding classes: sensible and sound presence rules with outputs for missed school; advanced interferences; focused interferences for students with long-lasting presence issues; and plans to improve commitment and personalization with the students and family units that may influence presence rates (Pinkus, 2008).

Academic achievement. Having literacy refers to the student's need to be proficient in a given subject (Pinkus, 2008; Sawchuck, 2010). Students do not have the required literacy to participate in grade-level work in many of their schools (Colby et al.,

2005; Lachat & Smith, 2005; Pinkus, 2008). When students become competent to attain the abilities required to progress, their ranks enhance, appearance raises, and attitude improves (Pinkus, 2008). Progressive high schools and their improvement models across the nation are engaged in investigating the individuals struggling and are involving youthful literateness advocacy as element of their game scheme (Pinkus, 2008).

Methods in establishing early warning data systems. According to Lachat and Smith (2005) as well as Pinkus (2008), administrators developing early warning systems need to be familiar with their ability to identify the students who are in danger of failing. In investigating variables that provided students approximately 75% probability of failure, Pinkus (2008) noted that not more than 79% of attendance rate appeared to be the tilting point in the eighth grade, whereas not more than 69% of attendance rate appeared to be the tilting point in ninth grade. To ensure that the signs as utilized contain the greatest probable output, instructors and managers utilizing early warning systems need to be equipped to manage their signs and triggers as innovative or favorable knowledge appears present (Pinkus, 2008). A former administrator, Ralph Thibodeaux of Abbeville (Louisiana) High School, reassessed the causes in his early warning data system, which allowed him to precisely investigate students who are in danger of failing (Lachat & Smith, 2005; Pinkus, 2008).

Bobby Franklin, who is an authorized worker at Louisiana Department of Education, argued his point of view and warned against the application of early warning signs because of the potential problems the program could cause for students (Pinkus, 2008). Furthermore, Pinkus (2008) reported that not only does an early warning system decrease an instructor's ability to recognize an emergency with reference to the students

in danger of failing but also, when the list is substantial, the instructors show incompetency to address the burden. Ultimately, few results impact decision-making around the application of early warning data (Pinkus, 2008). As design of greater output danger variables and the capacity for interference together deviate, districts and schools are required to confirm the data that appears to be actionable and prognostic for their specific system (Pinkus, 2008; Sawchuck, 2010).

Pinkus (2008) reported that there are numerous methods to acknowledge early warning signs and produce the data systems to locate them. He argued that this task may be finished through utilizing class systems that located student data for periods. He also reported that it may as well be finished with convenient study and instruments present at the school level. Pinkus (2008) illustrated in his research that one such method has been applied in Boston, Chicago, and Philadelphia. The focus of this method was to assess previous record on failure and to investigate general educational danger variables. Although, excluding exclusive longitudinal researches, simpler assesses may be performed utilizing previous transcript and school data (Pinkus, 2008).

According to a research conducted by Sawchuck (2010), in order to utilize graduation provisions and reverse engineering information, government schools in New York City produced metrics to gauge whether students are on track to graduation. According to Pinkus (2008), numerous states such as Massachusetts, Florida, and Texas are using these metrics. Pinkus (2008) suggested that these metrics are also used to analyze growers by involving the College Board and the National Assessment Governing Board, as well as entities like Innovative Visions and CCSR.

e-Leadership

A school administrator is a frontrunner in school administration, and his or her primary function is to supervise unceasing enhancement of the school culture, encouraging student learning (Chirichello, 2010). Conventional leadership refers to a procedure where a single individual of an entity impacts and administers the attitudes of additional members for achieving shared objectives (Burns, 1978). Chirichello (2010) contended leadership and academic workplaces are meaningful elements influencing the efficiency of the school. Furthermore, the administrator of the school needs to view the complete workplace and attempt to develop a strict association between its varied dimensions for assisting student progression and cultural unity. Chirichello (2010) noted that everything needs to be accomplished when school administrators attempt to define school procedures to encourage student learning and to improve student behavior.

Through comparison to conventional leadership, e-Leadership calls on the skill of an individual to impact the attitude of additional in an online technology-arbitrated culture (Chamakiotis & Panteli, 2011). A study conducted by Gurr (2004) suggested that the leader should institute procedures authorized by technology rather than through technology itself. Additionally, several studies applied the variables discussed before in multiple methods. For example, school technology leadership outlined in the study completed by Anderson and Dexter (2005) suggested the existence of a technology group. Dexter (2007) noted that technology leadership as administrators' engagement in technology oriented liabilities at schools presenting laptops in schoolrooms.

Divided leadership refers to a social division where the role of leadership is stretched over the task of a number of individuals, and the work is finished via the

communication of numerous leaders (Spillane et al., 2001). They also stressed that schools are often divided in leadership because of the many different situations in which different leaders have power. In comparison to virtual clusters, schools based in Israel are conventional workplaces with a graded assembly, where the notion of divided leadership appears to be barely useful to them (Spillane et al., 2001). Therefore, e-Leadership in academic institutions frequently shows the school administrator's leadership and deviates on a range from a controlling approach, for example top-down leadership, to a liability or delegating method (Tan & Aloysius, 2011).

In their study, Blau and Presser (2013) investigated e-Leadership by secondary-school principals through a school data management system called the Mashov School Management System, their promotion of e-Leadership by teaching staff, and the effect of e-Leadership on school effectiveness. They noted that the Mashov School Management System advocates liability, which is a combination of delegating and an order method together. According to them, every individual of the entity receives access to the database. School administrators get access to all pedagogical knowledge relevant to their school. Department chairs may observe the entire data relevant to their departments. Homeroom instructors get access to the knowledge pertaining to the behavior of their students in multiple classrooms. Students can contact their instructors; and parents can also contact their children's instructors.

A study conducted by Rogers (2003) explained the diffusion of innovation theory, and noted that diversity in the degree of welcoming innovative technologies is commonly divided into a bell curve. The bell curve shows a progression from modernizers (2.499%) and early recognizers (13.499%) to initial mainstream and late mainstream (33.99%) and

ultimately those who lag behind the adoption of new technologies (15.99%). On the grounds of this method, e-Leadership via a school management system deviates as per separate variations in innovation recognition among the school administrators (Rogers, 2003). Specifically, a study conducted by Avidov-Unger and Eshet-Alkalai (2011) established fundamental factors of execution: comprehensive novelty and isles of novelty. In the Isles of novelty model, the novelty is executed solely through a minor fraction of the entity and is commonly emphasized on a specific content domain or a specific job (Mioduser et al., 2006). Comprehensive novelty models, in comparison, promote the idea that the execution incorporates each degree of the entity, therefore developing an innovative organizational environment (Mioduser et al., 2006). By taking into notice the various studies, Avidov-Unger and Eshet-Alkalai (2011) determined that the supposition of achievement routinely spending from the isle of novelty to the rest of the entity is flawed. The isle of novelty is inaccessible from the entire workplace, leaving the primary decision making power to those in leadership positions (Mioduser et al., 2006).

Fuchs (1995) noted that the data system and electronic communication via the school are statistically significant only when multiple instructional staff are engaged in inputting data and utilizing the system frequently. Thus, when initiating the execution of a school data management system, an administrator needs to incorporate the procedures to all instructional staff. Furthermore, Fuchs (1995) suggested that the progressive execution of alterations in schools is impacted through each shareholder's input in the procedure needs, which not only involve instructional staff, but also students and their parents or guardians. Few systems associate student data pertaining to school attitude,

prospectus reserves, communication among employees, and school and home associations (Wayman, 2007). Blau and Presser (2013) gained experiential advocacy in extensive sample contrasts between the execution of the Mashov system, examined in the literature, and the execution of the system through instructors and family units.

Blau and Presser (2013) contended that online school information management systems (SIMS) have become an important tool for effective management of pedagogical information and student data. They defined e-Leadership as the capacity of a leader to engage others so they can work collaboratively in technology-based classrooms. According to Blau and Presser (2013), in order to enhance e-Leadership, school leaders should follow two important steps: first, they should mandate the use of SIMS among students, teachers, school leaders, and parents; second, they should assign e-Leadership tasks to school personnel and actively monitor its use at school (Blau & Presser, 2013).

Summary

In this chapter, the researcher has reviewed the literature. The methodology is presented in Chapter III. An epoche was included in chapter IV. Participants' individual stories are presented in Chapters V through XIV. Chapter XV includes a narrative analysis. The final chapter, Chapter XVI, discusses the Summary, Conclusions, Implications, and Recommendations for future study.

Chapter III

Methodology

The purpose of this qualitative phenomenological narrative study was to explore the perceptions of administrators regarding the implementation of a data dashboard system in one Texas Charter School System. This chapter outlines the following areas related to the research methodology: (a) purpose of the study and research questions, (b) research design, (c) the participants and the setting, (d) data collection, (e) treatment of the data, (f) provisions of trustworthiness, and (g) summary.

Purpose of the Study and Research Questions

The purpose of this qualitative phenomenological narrative study was to explore the perceptions of administrators regarding the implementation of a data dashboard system in one Texas Charter School System. The following research questions were used in this study:

1. What do school system administrators perceive as benefits to implementing a data dashboard system for administrators, teachers, parents, and students?
2. What do school system administrators perceive as ways that a data dashboard system supports data-informed decision-making?
3. What topics should the dashboard system encompass that are currently not being addressed?

Research Design

A qualitative phenomenological narrative study was used for this study. According to Creswell (2013), “Qualitative researchers use an emerging qualitative

approach to inquiry, the collection of data in a natural setting sensitive to the people and places under study, and data analysis that is inductive and establishes patterns or themes" (p.44). Creswell (2013) also indicated that the color of participants, the characteristics of the researcher, and the detailed analysis of the problem are main components of the ultimate product. In addition, Creswell defined phenomenological study as describing common meanings from lived experiences of individuals around a phenomenon. Creswell (2013) listed phenomenological research features and steps. The main data collection method for this study was face-to-face interviews. The participants were charter school administrators, campus principals, assistant principals, and area superintendents. The perceptions of charter school administrators regarding the implementation of a data dashboard were investigated using a qualitative phenomenological narrative design.

The participants and the setting. This study was conducted at campuses of a charter school system, which is the largest charter school system in Texas. 10 administrators from these campuses were selected via purposeful and convenient sampling. A purposeful sample was used to provide an in-depth understanding of the phenomenon. Creswell (2013) suggested that convenience sampling be used when the participants were available and willing to participate in the study.

Participants were selected from a pool of possible candidates based on an initial screening process. Specifically, participants met the following criteria:

1. The participant was a charter school administrator serving as a principal, assistant principal, or area superintendent.
2. Each administrator had two or more years of experience in the district and in their current position.

3. The participant was willing and able to share his or her experience about dashboard usage.

An invitation letter was sent to all school principals, assistant principals, and area superintendents who met the above criteria. The first 10 participants who met the above criteria, who were willing and able to share their experience, and who were readily accessible for face-to-face interviews, were selected. The participants were not totally representative of the charter school district population.

Data collection. The primary data collection for this study was face-to-face interviews. Creswell (2013) noted that qualitative researchers sketch a complex picture of the problem under study, report multiple standpoints, and identify the dynamics involved in a situation.

The researcher prepared a guided protocol for the research (See Appendix A). Prior to contacting participants, the researcher obtained permission from the charter school system's Institutional Review Board (IRB) and the Lamar IRB to conduct the study. Following approval from IRBs of both the charter school system and Lamar University, the researcher conducted interviews with the administrators.

Based on a protocol developed by Creswell (2013), the steps for interviewing included choosing research questions that focused on exploring the administrators' perspectives about data dashboard implementation. The researcher used the charter school system's administrator directory to find the contact information of school administrators. The researcher created an on-line link to the interview questions for the school administrators who wanted to participate in the study. The researcher sent an email to all administrators asking whether they wanted to participate in the study. The

email included a link to the interview questions and guided protocol. Contact information for both the researcher and the academic advisor were provided to ensure the participants of the authenticity of the study and to increase trustworthiness. Finally, the researcher determined the location for the interview, obtained consent from the interviewee, and used proven interview procedures (Creswell, 2013). The administrators who agreed to participate were contacted by the researcher for a time and place for the interview. The interview was optional and there was no incentive for participation and no consequence if they opted out. The first 10 participants who met the above criteria, were willing and able to share their experience, and who were readily available for face-to-face interviews were selected. Out of the 10 participants, seven were principals, two were assistant principals, and one was an area superintendent. Six of them were males and four of them were females. The interview required approximately 30 minutes to 45 minutes to complete all the protocol questions. After completing the interviews, the response data were reviewed for further analysis. Interview procedures, based on Creswell's (2013) work, were used throughout the study.

Treatment of the data. After interviews were recorded, the researcher utilized the suggested methods by Creswell (2013) for qualitative research analysis. According to Creswell (2013), this method includes: structuring the data, coding the data, and lastly depicting the data. Creswell (2013) explained that the process of coding involves aggregating the text or visual data into small categories of information, seeking evidence for the code from different databases being used in a study, and then assigning a label to the code. Once the interview recordings were collected, the recordings were transcribed. Then, they were coded and the researcher analyzed the responses to discover common

patterns within the perceptions of the administrators about the implementation of the data dashboards. After preliminary patterns were established, the responses were reread to verify the existence of those patterns. Upon the completion of these steps the researcher created a matrix, as suggested by Creswell (2013), to inform the reader of depth and forms of data for this phenomenological study.

Provisions of trustworthiness. According to Creswell (2013), "Validation in qualitative research is an attempt to assess the "accuracy" of the findings, as best described by the researchers and the participants" (p. 249). Creswell (2013) viewed validation as a process that is the distinct asset of qualitative research. He also noted that the interpretations of researchers on his/her past experiences, prejudices and predispositions are highly important. Creswell (2013) explained some techniques to ensure credibility. In this study, the collected data were tested for trustworthiness through investigator triangulation, prolonged engagement, persistent observation, and standardized system of data collection. The researcher also sought the opinion of others, and accepted criticism when needed, to draw out human perceptions as well as increase the validity and credibility of the research (Creswell, 2013).

Summary

The methodology was presented in Chapter III. An epoche was included in chapter IV. Participants' individual stories are presented in Chapters V through XIV. Chapter XV includes a narrative analysis. The final chapter, Chapter XVI, discusses the Summary, Conclusions, Implications, and Recommendations for future study.

Chapter IV

Epoche

According to Creswell (2013), the process of creating an epoche allows for readers to better understand the researcher's personal experiences and how they might affect the study. When I was a principal, I always wanted to have some tools to help me identify teachers' performance and why some students were not mastering specific concepts. I wanted a system that would provide real-time information to users, including students, through customized user-friendly dashboards. I wanted a dashboard system that would provide all of our stakeholders with the critical information they needed to effectively interpret and act on data in order to keep them on track to student, school, and district-wide goals. By itself, data has no value. When data is put into a form that is easily understandable, it becomes information. When information is used to guide the decisions in the best interest of the students and families we serve, it becomes knowledge (West, 2012).

Educational technologies are the future in all areas. Thirty years ago, there were no computers in schools or in most houses. But today we carry computers in our pockets. As educational leaders, we need to focus on defining what technology information we need in the future. The question is what measurements or developments are needed to set the needed future leadership. I work at a charter school system, a network of high-performing K-12 public charter schools across Texas, that focus on providing science, computer technologies, engineering, and math education (STEM) to traditionally underserved students.

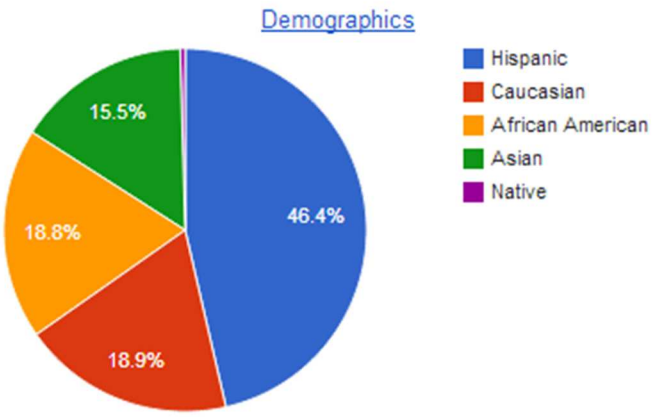

Figure 1. The student demographics at this Charter School System.

The 30,500 students in this charter school system are diverse: 61% receive free or reduced price lunches, 80%+ are nonwhite (49.7% Hispanic, 19.7% African American, and 13.2% Asian, Figure 1). The charter school system demonstrates that with a rigorous, high-quality program and the right social and emotional supports in place, all students, regardless of racial or economic background, can achieve outstanding results.

The schools in this charter school system consistently outperform regional and state averages in all four core subject areas and are making great progress in closing the achievement gap for minority and low-income students. Because these schools have very a diverse population, the leaders, principals, and area superintendents should have global leadership skills and a global mindset to effectively influence current schools.

This charter school system recently applied and received a Race to the Top-District grant to implement initiatives such as a professional teacher-principal development program and evaluation system, customized-individualize learning, PBL learning, and a data dashboard. This charter school system has the foundation and momentum to successfully carry out a major improvement initiative. This charter school system uses Race to the Top-District to further personalize the way each student uses

time, receives support to master essential skills, and deepens understanding of content. The schools achieve this by redesigning their school day, expanding their approach to inquiry-based teaching and learning, and upgrading their data infrastructure, resulting in a model for personalized education that can be replicated in the nation's schools.

So, the data dashboard initiative with the help of the RTT-D grant helps school leaders in this purpose. The wealth of information in the dashboard and the synthesis of that information on the customized dashboard significantly strengthen teachers' ability to determine the most appropriate learning recommendations for every student. Students have anytime access to their learning profiles, which are instantly updated whenever they have taken a quiz, completed a problem set, submitted an assignment and received a grade or comments. Teachers use this information to determine personalized learning recommendations. This dashboard also provides all of our stakeholders with the critical information they need to effectively interpret and act on data in order to keep them on track to student, school, and district-wide goals. The dashboard, with its easy access, also provides transparency among all stakeholders. Improving the existing data systems by building out the charter districts' Data System and developing customized data dashboards to provide real-time data to inform the first two strategies (and our system more broadly), and to support students in setting goals and creating personal learning plans are main goals.

Before the implementation of the data dashboard system, principals wanted data dashboards to see their schools' performance all at once, and then to be able to drill down in specific areas of concern. They wanted tools to support deep dives into teachers' performance in specific areas of content mastery and tools to help identify why students

were not mastering specific concepts. Principals also wanted more technology in the classrooms to support differentiation and address unique needs of individual children, and they wanted a specific person on campus to coordinate instructional technology.

With this data dashboard creation and implementation, I took some risks because the outcome or the way the information presented might be criticized for some stakeholders. However, it solved many problems by accessing the most critical data easily. I expected that there were other ways or tools available to get access to information and there were some disadvantages because it would be a tool available online and would require internet access from a computer/tablet or smartphone which not every person has access to. However, I found the project very innovative. It fits into 21st century life in which even toddlers have access to internet on iPads, iPhones, Android devices etc. I collaborated with my colleagues to get their feedback and connect, synthesize, and transform the concepts and ideas related to global leadership and key traits.

With this project, all stakeholders have access to data, so they have a chance to expand the circle of ownership and accountability for solutions across various kinds of boundaries, and in so doing support the development of future leaders who may have very different backgrounds and styles. Transparency is very important when working with other people. Leaders need to have clarity in their actions.

As leaders in the central office, everyone should better understand what best strategy could be used to successfully lead teachers and administrators in those areas that are different than in the Houston Area where the central office is located. To design a data dashboard, first the findings about culture can help leaders understand their own

cultural biases and preferences. Understanding their own preferences is the first step in understanding that people in other cultures might have different preferences. Second, the findings help leaders comprehend what it means to be a good leader. Different cultures have different ideas about what they want from their leaders. Therefore, by doing so, leaders can adapt their style to be more effective in different cultural settings. Finally, global leaders can communicate more effectively across cultural and geographical boundaries. By understanding cultural differences, leaders can become more emphatic and accurate in their communication with others.

This study provided all stakeholders with the critical information they need to effectively interpret and act on data in order to keep us on track to student, school, and LEA-wide goals. Data-driven strategies must be embedded in the culture of schools and school districts. Web-based business tools can help school leaders make sense of critical information and get a better handle on the key measures that drive student achievement (Carr, 2008).

The charter school administrators were initially interviewed before the building of this data dashboard system. Most of them said that having more adults focused on looking at the data and reaching out to students in need makes a difference. They added that data dashboards could help them to drive immediate decisions. According to them, the data dashboard project needs to address the following major activities associated with this project.

This dashboard ensured that all stakeholders have real-time, actionable data to keep students and schools on track to reach our ambitious goals. Below is a summary of the initial findings from those stakeholder interviews:

- Teachers wanted to use technology to scaffold support for each student and differentiate to meet unique learner needs. They wanted more technology in the classroom, especially hand-held devices, and training on how to use any new technology. They also expressed the desire for course offerings at the middle school aligned with the high school courses to better prepare students at an earlier age.
- Parents wanted easy access to lessons and classroom activities, as well as improved online resources to track their child's goals and performance. Parents also want rapid notification if their child is struggling.
- Students wanted access to technology in the classroom and beyond school hours and wanted technology as an enabler to make the topics they are learning in the classroom relevant to them (helping identify real-world applications).
- Principals wanted data dashboards to see their schools' performance all at once, and then to be able to drill down in specific areas of concern. They wanted tools to support deep dives into teachers' performance in specific areas of content mastery and tools to help identify why students are not mastering specific concepts. Principals also wanted more technology in the classrooms to support differentiation and address unique needs of individual children, and they want a specific person on campus to coordinate instructional technology

One key component of the data dashboard is a user-friendly dashboard, linked to a charter's data system and customized to end users, including students. The dashboard

draws from all of the district's databases and show all student assessment metrics, in addition to other student data such as attendance. Using the dashboard, students are able to monitor their progress toward college- and career-readiness in real time, and engage in data-informed conversations about their progress with teachers and parents. Seeing their own progress on a regular basis fosters students' sense of efficiency and self-confidence because they literally see how their efforts result in progress toward their goals. Being able to monitor their progress allows all students to:

- Be able to be involved in deep learning experiences in areas of academic interest and have access and exposure to diverse cultures, contexts, and perspectives that motivate and deepen individual student learning
- Master critical academic content and develop skills and traits such as goal-setting, teamwork, perseverance, critical thinking, communication, creativity, and problem-solving

The data dashboard project has been finalized and the charter's administrators have been using it for several months. The data dashboard provides data to school and central office administrators. With this study, I interviewed administrators to explore their perceptions on implementation of this data dashboard project. During interviews, I felt that they were part of the project for the purpose of expanding ownership and getting results through relationships. Through interviews, I learned more about the project's counterparts and their mindsets so that I can see the differences. We all stated that the dashboard project serves the best interest of the students. I acknowledge that as the researcher and a school administrator, I have interests, feelings, and opinions that influenced my decision to study this topic. Bracketing is essential to limiting researcher-biased conclusions to research

data which should only consider the perspectives of the participants (Creswell, 2013). I have set aside my personal bias to study the issue from the perspective of the participants to the best of my ability.

Chapter V

We Did Not Have Such a Dashboard

Mr. Sean is a talented educator and currently the principal of a charter school in Texas. He began his career as a computer teacher and later became a charter school assistant principal, and has now been a principal for several years. During his career, Mr. Sean changed jobs and moved often. In all, he has spent a total of 18 years in education. Seven years were spent as a teacher and 11 years in an administrator role.

Benefits to Implementing a Data Dashboard System

When I asked him about the benefits of implementing a data dashboard system for administrators, teachers, parents, and students, Mr. Sean told me that while he was a principal in another state, he had monthly meetings with teachers to discuss students' test results and evaluate their progress. They did not have a data dashboard. He explained:

> We were getting all data by hand: records from the dean of academics, lesson plans, etc. We were spending a significant amount of time getting those data. Having access to a data dashboard would give administrators and teachers the ability to get quick access to data whenever they needed it.

Internal needs. Having access to a data dashboard impacts schools. I asked Mr. Sean about the internal needs for creating a data dashboard. Mr. Sean emphasized the importance of daily data use at the schools and described how and why he uses data. He remarked:

> I and my staff members always need access to data to understand how students are doing, and what they could do to improve their knowledge. We need to look at test scores, attendance, and discipline records to arrange tutorial groups to

> increase the students' learning. There is so much data. Harvesting it for school use, students' use, and teachers' use is very important. Without a data dashboard system available, I do not see many people making use of the data.

External needs. A data dashboard system can also assist in meeting the needs of external customers, such as parents. When I asked him how a data dashboard could benefit the school with respect to its outside customers, such as parents and community members, Mr. Sean explained:

> Parents, community members, and stakeholders want to know what is going on in the schools. Parents especially want to know about their own children. Delivering the information they need to know day-by-day and meeting their needs increases their involvement in school education, and this will increase the learning of all students. Community members may need access to the school profile to check how the school is doing. They may also want to take ownership of the school if data is provided them in a simple way. If some school data such as school attendance, school demographics, and school test scores are available on the school website with numbers and graphic representations, it will help them understand the school.

Interpreting the data. When I asked Mr. Sean how he interprets the data when he has access to it, he said that having a data dashboard would help everyone analyze data very quickly and enable them to address student needs immediately. He reviews the data with staff members and studies trends to determine whether they are seeing growth or a decline in numbers.

Ways a Data Dashboard System Supports Data-Informed Decision-Making

Mr. Sean feels that a data dashboard would provide school administrators and teachers with "an excellent opportunity [to make] informed decisions." I asked him how he could use a data dashboard system in conversations with parents and staff. He described it this way:

> During monthly parent-teacher conferences, we could review students' records easily. At one of the schools I supervised previously, we had reading specialists and math specialists guiding teachers to make better decisions. But without a data dashboard, this is a time-consuming task for everyone.

Monitoring the progress. Data dashboard systems can also assist monitoring progress. I asked Mr. Sean how he uses data to identify persistent problems and monitor the progress of a school over time. His answer below is based on his past experience at previous schools:

> We were having meetings to look at test scores, attendance records, college acceptance rates, SAT/ACT scores, and discussing any problems, checking the progress. We were using buckle down benchmark assessments and arranging tutorials based on these scores. After 4-5 weeks, we were giving another benchmark test to compare the progress that students were making and arranging tutorial groups accordingly.

Accessing the data. Data dashboards provide several options for accessing the data. Mr. Sean told me that parents at his school have access to a mobile system where they can see their child's day-to-day grades. He went on to say:

> They check homework assignments. They can email teachers from the

application. They can see attendance records, discipline records. Also, they can check school news on this parent mobile application. Moreover, they can check the school calendar, and they can even make payments for school lunches. They can also see if there was a mock test given in the test center section and check the results.

On the other hand, the parents also need to access school data overall, such as test scores, attendance rates, financial rates, etc. Even though most of these data are available on Texas Education Agency (TEA)'s website, a simple data dashboard would make parents more informed, because not all parents can find this data easily on TEA's website.

Recommendations for System Improvements

In closing my conversation with Mr. Sean, I asked him if he had any recommendations regarding system improvements. In explaining his response to "How could a data dashboard be used to support evaluation and to provide cost information," he suggested the following:

> Evaluating teachers, students, and administrators is very important because it tells them what is working and what is not. If data is not readily available, it does not get used. Without that data, it is difficult to determine and compare progress. The availability of a data dashboard would facilitate evaluation discussions. A data dashboard allows the administrators to talk to teachers about how they are performing and where they need to improve.

Cost information. I asked Mr. Sean how a data dashboard could be used to support evaluation and provide cost information. He feels that having the data in one place is very useful. He said:

> School administrators should benefit more by developing a data dashboard. Currently, there is so much software being used to meet the needs of academic, finance, HR, and operations departments. Collecting data from each of these software programs and presenting them will be costly. But, I think, it will bring more benefits, and it will help us to make right decisions.

Sharing the information. Some of the data in data dashboard systems can provide valuable information to the public. I asked Mr. Sean if a data dashboard could be used to share information with public. He had this to say:

> The public wonders how much we do spend, and how we perform academically. Everyone wonders about this. Having access to these data publicly somewhere on our website will help them understand our school.

Chapter VI

I Enjoy Having a Data Dashboard System

Mr. White has been in education for 13 years and is currently serving as the area superintendent for one charter school system in Texas. He began his career as a computer science teacher, later became a charter school assistant principal, and then a principal for several years before taking his current job. In talking with him it was evident that he enjoys being part of education and wants to continue serving students and the community for the rest his career.

Benefits to Implementing a Data Dashboard System

I began my conversation with Mr. White by asking whether he thought there were benefits to implementing a data dashboard system for administrators, teachers, parents, and students. He thinks data is "crucial" for the education system, for administrators, teachers, parents, and students. He explained:

> It is how we measure student progress and check completion of a task. I believe a data dashboard is very much needed and we make use of our current dashboard system for everything. For example, when we talk about attendance rates, we want to know if we are meeting our attendance rating goals. Or are we not reaching our goals? We can see daily attendance rates on the dashboard, that's perfect. Then, when we have conversations with our principals and other administrators, we look at the data and ask ourselves, "Is it appropriate? Is it good enough? Or do we need to improve it, and how?" That gives us a chance to get to know about our current performance plus talk about plans and programs. It is very important for school administrators, for everything such as attendance, student

> academic success, and enrollment numbers, etc. I can probably list more and more things.
>
> Regarding parents and students, it is needed too because parents are a part of the education system. We need their involvement in education to serve our students and data dashboards help them see their students' progress online very quickly. They can know their students' attendance, academic performance, test scores, discipline, anything they are interested in seeing. They can access this information online, and a data dashboard provides them a great deal of information. It helps them understand how their students are doing, and it helps us involve our parents in education. It has a similar benefit for students. They can see their progress, their current performance. It helps them understand where they are and what they need to do to improve. So, I enjoy having a data dashboard system.

Internal needs. Having access to a data dashboard impacts schools. When I asked Mr. White about the internal needs for creating a data dashboard, he told me his schools have many sources of data. He went on to say:

> [The data] comes from different sources, such as Eduphoria, Naviance, Student Information Systems, etc. Finding data in each platform and trying to analyze it is very challenging. So, the data dashboard comes in handy and helps us see all the data in one place.

External needs. With respect to the external needs, such as parents, Mr. White felt there were several ways in which a data dashboard could be beneficial. He described some of those ways as follows:

> I believe it helps us involve our parents in their students' education. We can have conversations about the data available on the dashboard with parents and they're able to see their students' progress. It helps us create conversations with parents on objective data. Also, it helps them understand their students' progress and current performance. So, it gives us a chance to get together as a community and just focus on our students learning. I think it's a great deal.

Interpreting the data. Mr. White told me that principals and administrators can use the data they access on the data dashboard to be proactive. He explained:

> When my school principals and their administrators review the data, they generally look for some flags. Are there any students falling behind? Or is there an enrollment issue due to lack of applications? With the help of data, we sit down together and make plans to prevent any issues.

Ways a Data Dashboard System Supports Data-Informed Decision-Making.

I continued my conversation with Mr. White by asking about ways a data dashboard system supports data-informed decision-making and how he could use it in conversation with parents and staff. He described several ways data informs decision-making:

> We use [the] data dashboard to make data-informed decisions. For internal campus communications with our teachers and administrators, we use the data dashboard all the time. For example, when we meet with our principals, we go to the data dashboard and look at the enrollment numbers. "What is the enrollment number today? What was our goal? And what is the difference now? What should we do if enrollment numbers are below our targets?" So, the data dashboard helps

> us to have access to crucial data immediately, and we can start a conversation with the principals. It is similar with student academic performance, attendance, and other things. It is also necessary for parent-teacher conversations and parent-admin discussions. Data dashboard gives us a base on which to talk about their students. Everyone can check almost everything about a student. It helps us create the boundaries of the conversations and just focus on student learning. We have very positive experiences with data dashboard, and we have been using this for all of our parent conversations.

Monitoring the progress. Data dashboard systems can also assist monitoring progress. Mr. White explained that he uses data to identify persistent problems and monitor the progress of a school over time. He explained it this way:

> We look at data all the time, almost every day for school purposes. And we review it with our admin team on a regular basis. And then the schools review their data with their admin team and with their teachers. We find out the goals and the gaps, areas of improvement, and we talk about those and try to make improvement. So, it is very much needed and we have been using it a great deal.

Accessing the data. Data dashboards provide several options for accessing the data. I asked Mr. White about the data parents can access. He responded with the following:

> Parents have access to their students' information via data dashboard. They can check attendance, grades for each class, discipline records, lunch balances, and tests scores. So, they do have access to data they need, and this helps us have those conversations with parents.

Recommendations for System Improvements

In closing my conversation with Mr. White, I asked him if he had any recommendations regarding system improvements. In explaining his response to "How could a data dashboard be used to support evaluation and to provide cost information," he had the following suggestions:

> I believe the current data dashboard system is great. We have been using it for a while and are very happy about it. But of course, there could be some improvements. Schools use different software for different areas. For example, for teacher evaluation, we are using company based software to enter all the data. It can somehow be transferred to our current data dashboard system.

Cost information. I asked Mr. White how a data dashboard could be used to support evaluation and provide cost information. Mr. White also felt the data dashboard could be useful in monitoring costs. He said:

> We could transfer financial data so that we can check expenses and spending amounts per student. We need to start thinking about everything we do at school and every data we use, and we can definitely improve it. As administrators, we all want to make the right decisions. We all want to be able to monitor the progress with a data dashboard system.

Sharing the information. Some of the data in data dashboard systems can provide valuable information to the public. I asked Mr. White if a data dashboard could be used to share information with the public. He told me data is also beneficial for the public because:

Pulling the right data from the dashboard, narrowing it to what the public needs, and showing and providing this information to the public is very helpful for our school. This way, the public gets to know our school, gets to know what our numbers are, our statistics, and our performance. We are a part of the community. So, they need to know who we are, what we are doing, and what our performance looks like. I can also say that it is very critical for us to be transparent and share the information with the public.

Chapter VII

The Data Dashboard Provides All Related Data on One Platform

Mr. Scott has been in education for eleven years, and he currently serves as the principal at one charter school in the Texas system. He began his career as a science teacher and later served as a charter school assistant principal for several years. Mr. Scott has been the principal of a charter school for the past five years. Education is his passion and he likes being with students and seeing them grow. Working with students and parents is a rewarding and joyful experience

Benefits to Implementing a Data Dashboard System

I began my conversation with Mr. Scott by asking him his perception of the benefits of implementing a data dashboard system for administrators, teachers, parents, and students. He replied:

> Nowadays, one of the most significant problems for the schools is to be able to identify the strengths and weaknesses of a school system. Without data, it is not possible to identify them correctly. There are so many resources we use to access different data. Sometimes, it is tough for administrators to collect data from various platforms and then see the whole picture. It is also difficult for parents to have access to the different databases to get different information about their students and they do not want to do that. They wish to go to one platform to access all the data related to their students. Having access on one dashboard is also an expectation for teachers and the administrators. In this regard, the data dashboard provides all related data on one platform, so we can easily see it and compare.

Internal needs. Having access to a data dashboard impacts schools. As for the internal need for creating a data dashboard, Mr. Scott had this to say:

> The dashboard gives us the most critical data when we are looking for students who need intervention, and we make our plans based on the needs of our students. Besides, the data dashboard helps us track attendance, enrollment, and discipline data quickly in one place. This is critical, and with the support of the data, we can make our decisions faster and easier. We can go back to the data from this year and previous years, and we can see all of it on one screen. We can easily see the progress, even a change in the demographics or a change in the needs of the students.

External needs. A data dashboard system can also assist in meeting the needs of external customers, such as parents. In response to my asking about the external needs for creating a data dashboard, Mr. Scott said:

> The parents want to see what is going on in the school with their students. Currently, there are many educational discussions going on, not only statewide but also nationwide, regarding student progress. Because of this, parents want to see their student's progress throughout their educational life. They are looking for reliable data to identify and support their students to prepare a proper pathway to reach their goals. So, they are asking for tools from schools and from administrators to see their student's progress. In this regard, the data dashboard helps a lot.

Interpreting the data. Interpreting the data is an important aspect to implementing a useful system. When I asked Mr. Scott how he interprets the data when

he has access to it, his response was as follows:

> As administrators, we always make a comparison from year to year. We can use that to predict our students' performance at the end of the year because we are using practice star test. We use a map test at the beginning of the year, the middle of the year and we know the correlation between each test. So when I see this year's beginning of the year test, I see where my students are and what I need to do to make them ready for the star test at the end of the year or what kind of growth they are going to show according to the data because I can use it to compare it to the previous year's data. So that makes it easy for me to plan and predict what will happen at the end of the year. It also makes me comfortable. With the assistance of the data dashboard, we can predict our students' academic performance at the end of the year.

Ways a Data Dashboard System Supports Data-Informed Decision-Making

I continued our conversation by asking Mr. Scott ways in which a data dashboard system supports data-informed decision making, and how he could use it in conversations with parents and staff. He said:

> As I said earlier, before the data dashboard, we were trying to get different data from different resources, and it was not that easy. With the dashboard right now, we can see the whole picture easier. It helps us more easily compare the data, and we can see the needs and plan accordingly. The data dashboard also helps us see what is working and what is not working.
>
> We can also detect discrepancies easily to track accurate data. Currently, we are in the enrollment period, and I can easily use the data dashboard to see

> how many of our students will return next year, and how many new applications our campus receives on a daily basis. This helps me make a decision about whether to advertise our school more, if the numbers are less than what is expected. When we talk about the needs of our students, it is impossible to make a decision without any data… And when we talk about a lot of data, the dashboard makes it very easy to make a decision. With quick data, I can quickly have meaningful conversations with my teachers and parents.
>
> This is the same for the teachers. With the support of the data, they can easily see what is working and what is not working. Everyone makes meaningful data-driven decisions with the help of a data dashboard. Currently, parents have access to the academic grades of their students. They also have access to attendance records, lunch balances, library check-in and check-outs, discipline records, and test results. So, what else I am looking for?

Monitoring the progress. Data dashboard systems can also assist monitoring progress. I asked Mr. Scott how he uses data to identify persistent problems and monitor the progress of the school over time. He said:

> We have the data year after year. So when you put the data next to each other you can see what is going on because one year you have the beginning of the year data, middle of the year, end of the year data and you use the same programs and can see the progress of the students. And the next year if you are using the programs, the same resources, and the same implementation. So when you have the same starting point you can use it to predict, and if you have a couple years' data, you have better results and predictions and can use those to reduce errors.

Accessing the data. Data dashboards provide several options for accessing the data. Regarding what school data parents have ready to access to, Mr. Scott said that:

> Parents have access to the grades of their students. They have access to attendance records, lunch balances, library check-in and check-outs, discipline records, and test results. So, what else I am looking for? We also have different online resources for our students in school and after school and each of these resources have their own resources for the parents. The parents had to go to each of them individually to see the progress of all their students. Maybe if you combined them in one resource, the parents could easily track them. They wouldn't have to log in to see how many books their student read from, let's say, Excellent Reader, or how many questions they solved on Istation or on Reasoning Mind.

Recommendations for System Improvements

In closing my conversation with Mr. Scott, I asked him if he had any recommendations regarding system improvements and how a data dashboard could be used to support evaluation and provide cost information. He had these suggestions:

> Right now, we are using different software for curriculum tracking to see where a teacher is as far as instruction and progress. We do not have any curriculum data transferred to the data dashboard. We have to go to a different platform to check it. It would be nice to see this information on the data dashboard because of its easy access. I prefer to have access to all software from one location. When I do classroom observations, this new feature could help me see the teachers' progress in math, ELA or science, etc. I can see whether the instruction correlates with the

> data or not. This makes my observations meaningful, and I can easily discuss them with the teacher.

Cost information. In wrapping up our conversation, I asked Mr. Scott how a data dashboard could be used to support evaluation and provide cost information. He replied as follows:

> It would also be great if our financial expenditures were shown on the data dashboard. That way I could track how much funding our school gets from the state, and where the money is being spent.

Sharing the information. Some of the data in data dashboard systems can provide valuable information to the public. I asked Mr. Scott if a data dashboard could be used to share information with the public. He said:

> We could also share some school data with the public. They could check our school demographics, student-teacher ratio, test results, etc. with easy access to our data dashboard via the school website. You can use it to compare the progress, and that is very reliable data because you are keeping very accurate records and when you share the data with them . . . you can use it to provide a whole picture of the school so they can use it to answer questions or provide the knowledge or the information they are looking for in really compiled data. For example, during the open houses, the parents there ask questions related to demographics, related to academic performance, and related to discipline. So, when we say, "Okay, this is what happened the previous year, or the year before that, or the year before," we can use it to convince them about the true picture of our schools.

Chapter VIII

Dashboard Helps Us Work Smarter

Ms. Thomas has been in education for seven years and is currently serving as the dean of academics for kindergarten through seventh grade for a charter school system in Texas. She is also the testing coordinator and the special programs coordinator for this program. Ms. Thomas has a master's degree in education and administration. She describes herself as someone who is personally very interested and passionate about the curriculum aspect of education. Also, she enjoys being an advocate for teachers. In talking about her interest in education, she noted, "I want to help make sure that my school's curriculum is both the best curriculum for its students and one that the teachers can implement easily."

Benefits to Implementing a Data Dashboard System

I began my conversation with Ms. Thomas by asking about the perceived benefits of implementing a data dashboard system for administrators, teachers, parents, and students. She replied:

> I think the data dashboard is really good because it puts everything right there for everybody to see. It would make it easier for us to see all the information that we need. And also for parents, when they can see it, it makes us very, I guess, public. We are not trying to hide anything because we are showing them these are our numbers; this is what we are working towards. And it also keeps us in touch, I think, as it is updated on what our needs are on campus and what our strengths are. So it helps us determine what we need to work on.

Internal needs. Having access to a data dashboard impacts schools. I asked Ms. Thomas about the internal needs for creating a data dashboard. She responded:

> Having access to data dashboard would help us to have all the data there instead of having to look for it. Data is very important. We use it for a multitude of things, and it can make our jobs easier when we have all the data in front of us versus having to look at it or create the data ourselves by using spreadsheets. And then there is data we use to relate to our teachers, what our problem areas are and how can we address those needs. So having data, like I said, easily accessed, makes our jobs easier.

External needs. A data dashboard system can also assist in meeting the needs of external customers, such as parents. In responding to external needs for creating a data dashboard, Ms. Thomas said:

> When we have access to data that we can relate to our parents, we use it in our CNA and CIP meetings for kids so that we can create a better school system. Our goal is always to improve from the previous years. And so when we have access to this data, it helps us determine, what do we need to do? And how can we do that better? How can we make it better? There is always room for improvement. And having the data there shows our stakeholders that we are invested in our students' education; and we are always looking for ways to grow and make ourselves better.

Interpreting the data. Interpreting the data is an important aspect to implementing a useful system. When I asked Ms. Thomas how she interprets the data when she has access to it, her response was as follows:

> When we have all of our data together, we see whether there is a trend. For example, if we are having trouble in third grade, is it just third grade math or is it third grade math and reading? Then if it is the teachers or the students. So we look at the data to determine what our issues are. We also have created with our data a data room where we can share that with the teachers so that they can see the students that are in need of help. So having all that data helps us to continue to see what areas we need help in.

Ways a Data Dashboard System Supports Data-Informed Decision-Making

I continued my conversation with Ms. Thomas by asking about ways in which a data dashboard system supports data-informed decision making, and how she could use it in conversation with parents and staff. She said:

> Well, like I said, that data dashboard, when we have all of that there, we can get information quicker and faster. It doesn't take us so much time when we're having to go and look for the data. So all the data helps us... We're a data-driven school, we use data for everything, for attendance, for testing, for enrollment. We'd look at different numbers. And so when we have all this in one place, it just makes it easier and helps us work. Instead of harder, it helps us work smarter.
>
> Well, with our parents, when we have our annual CIP meetings and our title one meetings, we review stuff like: How did we do on testing? How are we doing on attendance, our growth? So we use all this data that we have to relate to our parents so that they see... that we are a strong school and how we help make students stronger here on campus. With our staff, we review data, especially now with our intervention meetings that we have after each benchmark and after each

> mock test; we sit down and communicate the data from those testing results and relay it to them. We also talk to them about our attendance. We look at the data from our attendance, our growth, our trend. If it were missing a lot of students on what days, why is that? So we relay. We always are informing our teachers about what we get from the data that we have available to us.

Monitoring the progress. Data dashboard systems can also assist monitoring progress. I asked Ms. Thomas how she uses data to identify persistent problems and monitor progress of school over time. She replied:

> We created a data room using information that we got from the dashboard, different areas, and we looked at the student growth from one test to the other; and we also look at their growth from previous years. If we have a student that did well, let's say, in fourth grade, but all of a sudden [in] fifth grade they're struggling, we try to figure out what happened. Is there an issue going? But we also look at how much students have grown from one year to the next. We also use it to monitor our ESL and our special education students. Are they growing academically? Or [have] they plateaued? Are they staying the same? And then we identify how we can help them increase their scores. So the data we have is always being used to monitor our students to see how they're doing.

Accessing the data. Data dashboards provide several options for accessing the data. When I asked Ms. Thomas which school data that parents have ready to access to, she responded the following:

> With our parents, they're able to access, basically, our school report, which is from TA, and data on their own student, like how they perform on tests. I think it

> would be good for them to see our growth from year to year. They can see how our students are performing. And also maybe teacher performance - how teachers have done from year to year.

Recommendations for System Improvements

In closing my conversation with Ms. Thomas, I asked her if she had any recommendations regarding system improvements. In explaining her response to "How could a data dashboard be used to support evaluation and to provide cost information?" She suggested the following:

> I think one of the things that's missing is our at-risk numbers on the dashboard; they're not there. That is part of our CNA. Since I do both CNA and CIP, that would be really great to have on there... to see our at-risk population on the data dashboard.

Cost information. I asked Ms. Thomas how a data dashboard could be used to support evaluation and supply cost information. She had several ideas about how it could be used. She said:

> For cost information, you can see all the numbers directly there, and we can use it to see how much our campus has grown. Or if the campus is struggling, it would help to evaluate what needs to be done to improve that campus.
>
> We could actually see what our budget is for, let's say, our special programs, our materials budget; and we could see what we've spent over the year and what's available for purchasing. We know if we have funds available, we were able to see it right away.

Sharing the information. Some of the data in data dashboard systems can provide valuable information to the public. I asked Ms. Thomas if a data dashboard could be used to share information with the public. She said:

> For public share, if it didn't have any student names, they could see everything. We let them see what our attendance is. The dashboard is really, I think, user-friendly and very visually friendly so that they can see the numbers. I believe that it would be great sharing the information with the public. They would be able to see everything, our attendance, our student populations or your cell population, so that they can... see that we are a diverse school and that we do service these different populations of students.

Chapter IX

An Important Benefit of a Data Dashboard is Its "One Stop Shop" Feature

Ms. Hunter has been in education for eight years and currently serves as the assistant principal for a middle school. She is also the Science Technology Engineering and Math (STEM) coordinator for the campus. Prior to Ms. Hunter's education experience, she worked as a trainer in the corporate world, but realized she needed to be in education. She mentioned that she had Attention Deficit Hyperactivity Disorder (ADHD) and wanted to help kids while they are in school so that they do not end up like her as adults. As an educator, her interest is targeting kids she sees that resemble her so she can bring them out of their shells. Ms. Hunter thinks that using her experience with ADHD, showing compassion, and opening up the world for them, will help kids with ADHD and all kids.

Benefits to Implementing a Data Dashboard System

I began my conversation with Ms. Hunter by asking about the perceived benefits of implementing a data dashboard system for administrators, teachers, parents, and students. She feels an important benefit of a data dashboard is its "one stop shop" feature. She explained it this way:

> When you look at the dashboard, it gives you so much information in one place that you can grab it and kind of go. What I mean by that, you're able to retrieve the data very quickly. You can analyze it from the year before to now, instead of having to go to different programs, so that alleviates a lot of time. It's more efficient for the administrator. As far as the teachers, it helps them gain an understanding of what are they doing in the classroom. They can see where the

> kids seem to struggle the most and can target those areas. So every year they're reflecting and growing. It also helps parents and students see what the school is doing, what the tracking is. If I'm interested in that school, but that school has low ratings, well, then I don't want to take my child there. So the benefit is that your school is doing everything it can for student success and other people are going to see that and want to bring their child to your school. Then you're going to grow.

Internal needs. I asked Ms. Hunter about the internal needs for creating a data dashboard. She felt it was specifically helpful when meeting with teachers. She said:

> I think, internally, a data dashboard will help by keeping up with the success of the teachers so you have a little bit stronger data when you're meeting with those teachers. It also helps when we're just aggregating the data for those teachers so that they have that information readily available to them.

External needs. A data dashboard can also address external needs. Ms. Hunter felt that the dashboard would be good public relations for the school. She explained:

> [With a data dashboard] parents and the community can see the rankings of the school with testing, and they can look at how the teachers are performing. They can see all of that information. They can see all of the diversity that's at the school, and again, it'll show that there's no prejudice. Everyone's allowed; we're making everyone successful. You can see the drop-out rates are low. You can see that the graduation rates are high. Those are things that really entice parents and the community. Then you build a network with the business partners, and then you have more parents getting the word out about the school.

Interpreting the data. Data interpretation is an important aspect to implementing a useful system. When I asked Ms. Hunter how she interprets the data when she has access to it, her response was the following:

> For us, the way we really look at the data, one of the things that we do as administrators, and number one for me, is look at how many students passed and how many students failed. I don't necessarily look first at all the breakdowns; I look at the rate between the two. If it's a lower rate, failing students, I go back, take a look and ask, "Okay, why were these the hardest parts for the child to understand?" Then I help the teacher understand that, and then we put everything into the tool that we have and we aggregate it and break it down step-by-step so that they get their groups. But if they don't understand what assignment it is that the child is missing, then they're not really even going to be able to do the intervention correctly. I can give them sheets all day long, but they may not understand that.

Ways a Data Dashboard System Supports Data-Informed Decision-Making

I continued my conversation with Ms. Hunter by asking ways in which a data dashboard system supports data-informed decision making, and how she could use it in conversation with parents and staff. She felt the biggest advantage was having easily understood data to share with parents. She explained it this way:

> The one thing with the parents, when we're communicating with them, is having the dashboard so that you have the information. That's number one. Quick and readily available, but also understandable. Because again, I can't give that parent the same type of printout that I would give a teacher because they're definitely not

> going to understand it. So I think having a very simple quick breakdown to show parents where their child is actually struggling, and show whether it's an academic struggle or a behavioral struggle, is helpful. And I think sometimes when we're able to see things in black and white or see things in front of us and not just be told something, it actually has a stronger impact.

Monitoring progress. I asked Ms. Hunter how she uses data to identify persistent problems and monitor progress of a school over time. She uses the information to help both teachers and students. She said:

> When I start looking at data that shows a teacher is constantly struggling, I try to figure out why? Is it because they don't understand how to teach the content that they're teaching, or do they not have the right resources? So I look at the data and I determine how many students passed, how many students failed, and what was it they failed? How can I help this teacher figure that out? Once I help figure that out with the teacher, and I get them signed up for training, or get them hooked up with a mentor that can guide them with some strategies, I then ask myself, "What is it that the students are missing? What's the best way?" It's not always about just throwing a kid into tutoring. Sometimes it's about actually having a one-on-one with the child and letting the child tell you what's going on because their mind may be totally abstract from what we're expecting. So we need to gauge it that way, and that's how we need to take a look at all the data.

Accessing the data. Data dashboards provide several options for accessing the data. When I asked Ms. Hunter what school data parents have ready access to, she responded with the following:

> Right now, parents can go through the portal we have for them. They're able to see grades that their student or child currently has, so in each content area they will see an assignment and a grade. I do think that it would be beneficial for the parents to be able to see passing rates as a whole. Of course, not listing other students, but for that teacher, and for that subject, a passing rate as a whole. So that they can keep up with it, so they can say "Okay, is only my child struggling or is the whole group struggling?" And again, that helps the parent ask themselves, "Where does my child need to be, and what do I need to do as a parent to offer more support and help for my child? Or is it the school? And is that a concern I can bring up to the school?"

Recommendations for System Improvements

In closing my conversation with Ms. Hunter, I asked her if she had any recommendations regarding system improvements. Rather than suggestions for improvement, Ms. Hunter had several things to say about its benefits. She said:

> I think one of the good things about having the dashboard is that it has that information there, and it tracks teachers and it tracks from previous year to current year. It supports the evaluation process and keeping that documentation and showing the instructional support and having a place where it's very easy to access and go back instead of having to go to a folder or go back to a different person who did the observation the year before. And I think that it helps keep a little bit more consistency with helping the teacher perform and grow.

Cost Information. A data dashboard system can provide information about costs. I asked Ms. Hunter how that information might be used. She felt the information could help in allocating funds and in helping teachers. She explained it this way:

> In considering cost information, it would probably help with expenditures when you're looking at curriculum. Because again, if you're looking at that data and you're seeing that everyone's struggling, for example, with ELA and its teachers from 6th through 12th grades that are having this problem, you have to ask, is it the curriculum? So if it's the curriculum, "Okay, if this is enough data to show, let's try to see what we can do to bring in a new curriculum and kind of pilot it." And so you start testing.
>
> If it's not working, figure out how it's not working. Test out different things. I think that would be really beneficial to give us an idea about how much is left in our budget – is that something we can do? I'll take a look at teachers' salaries. These teachers are out-performing. So what can we do to show those teachers that we really appreciate them to keep them with us. We don't want them leaving and going somewhere else and us having to rebuild our teacher population along with taking a look at intervention efforts for our students.
>
> So if we're looking at the data, we're keeping up with everything. We will know if there is as much money as is needed to be spent in the intervention efforts because we're doing everything in the classroom already.

Sharing the information. Some of the data in the data dashboard systems can provide valuable information to the public. I asked Ms. Thomas if a data dashboard could

be used to share information with the community and whether that would be beneficial. She responded:

> I think with that, one of the big things is going over enrollment whether it's increasing or decreasing. A lot of times if you take a look at the schools around town, you think "Well, that sounds like a really interesting school," but you can never find out any information about them. So if you see a school is growing in population, they have a lot of the different pathways that are STEM related, they can bring those students in. They can also determine an area of growth and strategic development plans for the areas that you're deficient in because then you can compare yourself to other schools that are similar to yours. And so, you reflect, you reflect on everything, you go back and wonder, "Okay, how can I improve our school, our campus, so that we can be bigger and better every year?"

Chapter X

Implementing Data Dashboard System Allows for Collaboration

Ms. Bee has been in education for seven years and is currently the principal for a charter school system in Texas. Her school serves kindergarten through eighth grade. Ms. Bee began her career in education in 2009. She started out wanting to fulfill her passion for science and teaching science, and that is one of the things that she really loves about her current school. She describes herself as someone who wants to impact a variety of subjects and help teachers get to that point in their careers as well.

Benefits to Implementing a Data Dashboard System

Ms. Bee and I first spoke about how she viewed the benefits of implementing a data dashboard system for administrators, teachers, parents, and students. She feels collaboration between all parties involved in the school is important and explained it this way:

> Implementing the data dashboard system allows collaboration between teachers, parents, students, and administrators. It gives everybody the same data to look at. Granted, the data might look different [depending on] what you're looking at: the group wide level, a campus level, course level, grade level, and student level. But I feel that a lot of times the data we look at as administrators is sometimes different than the data teachers are looking at, just in the sense of how they pull it and what they have access to; it's different than what parents have and understand. And the same with the students. The dashboard would help us all have a common point to start from, look at, and collaborate on.

Internal needs. I asked Ms. Bee about the internal needs for creating a data dashboard. She felt the dashboard would offer a clearer way of sharing data and student progress on her campus. She said:

> As far as the internal needs go, we do have a need. [We could share] student participation information, and a variety of things with our parents. We've had numerous meetings with parents for attendance; parents know their kids have been absent on certain days, but when we sit down and talk about percentages of absences for the year, or the different rules regarding truancy and whatnot, which are things that show up very quickly in our system, [without a dashboard] parents don't have any way of accessing something in a visual way that explains that to them. I think the dashboard that has recently come out - I've looked at it quite a few times and it puts those, it seems to have the ability... I know the student one hasn't come out, but seems to have the ability to put it in a percentage for parents to kind of see, and it could even be a different color to let them know "Your kid is really close to being in contempt for that."

External needs. I asked Ms. Bee what external needs she thought a data dashboard might address. One feature she felt was very beneficial was the ability to access all the data in one place. She said:

> We of course have meetings, but I think [the data dashboard] puts [information] in that place where, if they can log in and see that, we can all be working together. I think that as far as staff goes, right now, we have several different places that we log into to pull information, and it can be time consuming to log in to all the different systems. The new database we've been looking at and seeing all the

different things that have been posted there has brought [together] several of those different things we would [otherwise] have to login or do multiple clicks to get things printed all to one centralized location. [With this system] we don't have to have so many clicks. You just find the report you want, and you click right on that report and then you can go more in depth instead of having to go through a series of logins to get to them. I think that's been incredibly helpful in my end.

I think our teachers would really enjoy that as well. For internal needs, I think that the system (I'm guessing we're talking about the one that's being implemented right now.) allows for feedbacks in it and I know I like that. When we were first surveyed about it, I didn't have very many opinions or ideas because I didn't have an understanding of what it was going to be like. But having seen it and actually being able to see some of the feedback other principals have put in there was really helpful. I was like "Oh, that's a great idea. That would be very helpful to have."

Interpreting the data. Interpreting data is an important aspect to implementing a useful system. I asked Ms. Bee about how she interprets the data when she has access to it. Her response was:

We start by exploring data in our admin meetings and we look at the highs, lows, averages and very basic things like that. From there we take the things we're concerned about. Maybe we see a concern in high, low, or average so we'll look more in depth at that, and maybe we have a teacher we're concerned about and we want to look more in depth into that and [with the data] we can do that. We usually do that with our AP academics, but even sometimes with our AP

> discipline. I would say our focus on our campus has mostly been data related to academics such as the tests and the assessments the students are taking to monitor for progress.
>
> We've actually made some changes. We noticed that students weren't progressing and we noticed that through a couple of our assessments, as well as through walk-throughs that we've been doing. Our intervention time with our students wasn't as effective as we needed it to be, so we rearranged the school schedule just a little bit and arranged for a more private, quiet time where intervention students would have a space to go to instead of being in a loud classroom where they were trying to listen to the person working with them while filtering out all the noise. That's helped a lot, and we've seen that in the data as well.

Ways a Data Dashboard System Supports Data-Informed Decision-Making

I continued my conversation with Ms. Bee by asking ways in which a data dashboard system supports data-informed decision making, and how she could use it in conversation with parents and staff. She felt it helped a lot and explained why:

> I know the sub questions on this are about parents and staff, and I feel that we as a school are going – actually just in general in this field of education – we have all these tools now for checking where students are and for easily identifying what standards are being worked on and which students should be in a group. It's not like the teacher has to go in and do their own assessment and then figure out their own groups. It's streamlined. However, I feel that we're missing a piece now and that it kind of excludes the parents. I feel that we, as a school, are constantly

> looking at these students in these intervention groups and their teachers are constantly looking at them, but that's the only message we're currently sending to parents. It's something we're going to work on. [The problem is] that their progress within those interventions is not necessarily being sent home to the parents. I don't know if it will be something maybe on quarterly basis, a monthly basis, a weekly basis. But it would be to let the parents know that these are the areas we've identified and we're going to work on with your child. Now if they change, we send them another letter that says that as well. And this is the interventions that they're on.

Monitoring the progress. Data dashboard systems can also assist in monitoring progress. I asked Ms. Bee how she uses data to identify persistent problems and monitor progress of school over time. She replied:

> We did send that initial letter, but as far as the follow-ups, it's like now they've mastered this, or we're still in progress on these. But when they see progress, or we've met, but no progress has been made so we're still working on it, it's really important for parents to be involved and understand. I feel when we end up having meetings with parents, especially when we're getting up closer into second semester and talking about possible retention and things like that, the parents are feeling like their students are in tutorials so they should be doing well. But that doesn't necessarily mean that they're doing all the things [they need to do]. And there might be a lot of factors. It might be that we need to define the terms to the family. But it also might be that the student is not putting in their full effort, and I think that if we're giving that update to the parents and giving them the

opportunity to be involved in progress, I think a lot of them would appreciate it.

Accessing the data. Data dashboards provide several options for accessing the data. When I asked Ms. Bee which school data parents have ready access to, she responded the following:

> Currently, parents through their portal have direct access online to test center data, although not all of our test results are on test center, but most of their data is in there. They have access to the grade book with grades in it from the teachers. They have access to Teacher/Student PRS and DPS points online as well, and to their attendance. But the way those are calculated are more of a numerical base which is fine, it's good to have a number. But a parent doesn't necessarily say, "Okay well, there's this many absences. There's this many days in the school year, that's this percentage of days." That's not really something I think that goes through a parent's mind. They just think, "Oh, I see they have some absences."
>
> I don't know if they always put two and two together about how that all works. But they do have access to it. They just don't necessarily have it in a great format to look at. Not that it's a bad format. It's just, for a parent that's doing a quick glance, it's not as accessible. It would be awesome if we could change our website into different languages on that system as well. We have a lot of Spanish speaking parents and it would just make them more comfortable in being a partner in their child's education, if that information was readily accessible to them in a language they understand.

Recommendations for System Improvements

In closing my conversation with Ms. Bee, I asked her if she had any recommendations regarding system improvements. Specifically, I asked her how a data dashboard could be used to support evaluation and provide cost information. She suggested the following:

> I think expanding the dashboard to include aspects of budgeting and other things would be helpful. The curriculum pacing guide we have, and which we currently have to log in to a separate database for to pull a report for our teachers, if that were all in one place it would be really helpful. It's a great tool to kind of just check on where things are at. I know that we pull it, and we tell teachers "We're seeing this. We're seeing your lesson plans through here, and talk to us a little bit about your plan to make up this gap," or whatnot. So I think that incorporating that will be super helpful for us.

Cost information. With respect to monitoring costs, Ms. Bee felt there were several ways a data dashboard could be useful. She especially thought it would help teachers better understand the budget. She said:

> Specifically, I think budgeting will be very helpful to have in there as well. I have been working towards trying to get a way to make our budget a little more understandable to our teachers so they feel as if they know what our campus expenses are, and understand when we have money, what we want to spend it on, the priorities. And involving our teachers, and mostly our department heads and leaders on campus, in making some of those decisions of prioritizing through our CNA and CIP meetings.

> But along with that, I think having some aspect of budgeting information would be nice. Even if we could just allocate a certain amount. We used to have certain amounts that were kind of set that were expected to go to science, or math, or English. That got handed to the campuses, which is wonderful as well, so that we could make those priorities ourselves, but it'd be nice if even that was something that [was accessible]. They could see why if we said, "You guys have $1,000 in your department. Obviously that's not going to be the whole academic funds that we have, but $1,000 of what you guys want to vote for in your department." And that would also give them this idea if they could... I don't know how that would be tracked. But I do think that sometimes it's a very hard system to understand the budget, especially at the teacher level. And if there's any way that we can share that, I think it would be great.

Sharing the information. Some of the data in data dashboard systems can provide valuable information to the public. I asked Ms. Bee if a data dashboard could be used to share information with the community. She had this to say:

> I would love to be able to choose certain [data] to have displayed on our website. Maybe not all the time, but if we ever had a report that shared something, if there was a box that we could click that would just put that report onto our website, such as... I don't know… I feel like I'd always want an attendance graph on our website saying, "We have whatever percentage students at school today." I think that it's an encourager to remind everybody to come to school. And it's generally a high percentage. We're usually 97, 98. And so I think it would be like, "Oh, I better go to school. Everybody else does." We also could have on there... I know

we have significant progress at the campus level from the Quarter 1 to the Quarter 2, and we posted that in our teachers' lounge; we wrote it on a whiteboard in there for the teachers to see, to say, "Great work. We did all this progress." I think that when there is good progress, that's an awesome thing to put on our website as well, because it's an actual measured [achievement]. It's real data. It's an actual measured amount of progress where we're seeing our students grow, and it's a way to celebrate that publicly.

Chapter XI

A Big Push for Us is Intervention

Mr. Rodriguez has been in education for ten years and currently serves as the principal for a charter school system in Texas. He started his career in education as a Social Studies teacher, and has also taught AP World History and AP Human Geography. Mr. Rodriguez values education and its potential to make a difference in a child's life. Moreover, he believes the can interact with parents to help them understand the value of education.

Benefits to Implementing a Data Dashboard System

I began my conversation with Mr. Rodriguez by asking about the perceived benefits of implementing a data dashboard system for administrators, teachers, parents, and students. He believes a real benefit is the consolidation of data. He said:

> Most school districts, and ours especially, are not generating the data source, so we collect a lot of information but it's from disparate providers. We provide a MAP test and we get to store information back from the state and scores for GT and special education. We have all of these various scores that we're collecting from people all over the place and the dashboard allowed us to centralize that information and get a broader perspective on what all of it means in aggregate rather than just individual pieces. That's what I've seen.
>
> As far as teacher use, it really should lead the decisions they're making in the classroom. A big push for us is intervention. I think that's the right thing to do, use the tools for intervention, but more importantly I think day-to-day instruction can be driven by [the data]. That's what we're seeing in the trends provided

> through a dashboard system where you have all the information at once. As far as parent involvement, I haven't seen an awful lot of parents utilize information that we get from the dashboard except through the teachers. Most parents are overwhelmed when you start to talk about industry things like testing, assessments, the measures of the intellectual abilities, and things like that.

Internal needs. I asked Mr. Rodriguez about the internal needs for creating a data dashboard. He felt there were several internal advantages to a data dashboard. He said:

> The dashboard really gives us a whole picture of where we're at in terms of where our student body is, but then we can also move down through layers and move into classes, and individual students, to see their performance. And I go back to tests a lot. There's attendance features that let us track trends over time, and help us identify areas where we're struggling, and where we can improve. The behavior elements are certainly useful. I really like the idea of being able to track individual student performance as far as any type of referral and GPS points.

External needs. Mr. Rodriguez felt that external needs could be addressed with a data dashboard. Specifically, he felt access could help with both public relations for the school and with enlisting community help. He explained it this way:

> I think the community could benefit if there were a more generic portal. I think [it would be good] if they could go to some access point, like our website, and get a broad overview that has customized information, that the average (outside the building) stakeholder might have. You could even market some of the public relations work that we do, so we could share information about how often we're in the community, and who we're reaching out to in the community. But we could

> also share needs, so if there were some way to develop a needs list using the data, and we could reach out to the community, I think the school would benefit from having the dashboard provide some type of need identification where the community could support us.

Interpreting the data. Data interpretation is an important aspect to implementing a useful system. When I asked Mr. Rodriguez how he interprets the data when he has access to it, his response was the following:

> We typically find the data from the dashboard useful for the MAP test in particular. So Eduphoria does a good job, and the dashboard brings all this together, but Eduphoria does a really good job of helping us identify student performance on mock STAARs, the actual STAAR, and benchmark testing. But MAP is sort of a different animal, and from the school side, we don't have access to the background batch data. The best we can do is run largely PDF reports that lack some of the deeper information you can get. But the dashboard allows us to pull data that's formatted a little differently so we can apply it in a more appropriate way, I think. So what we do is take the MAP data from the dashboard as a baseline for students and that helps us begin to identify student needs. Then we overlay benchmarks, and then the STAAR data. And we start to really funnel down the needs of each kid in the classroom. And I know that the dashboard, by having all that information centralized, helps us quite a bit.
>
> Attendance is really a far simpler thing to interpret. You look at your attendance rate and it helps you see how healthy the budget's going to be and that directly translates to instructional resources that we can get into the classroom. It's

good to know if we dip below that 90% attendance. We want to be at 97%, but if we dip below 90%, we can apply for a waiver from the state, and the dashboard really simplifies that. It gives us real time data. I don't have to wait for an attendance clerk to understand that that's happened. I get a lot quicker notification that we should move ahead with a request for a waiver. And sometimes that'll happen on holidays that we're unaware of for large populations of minority students that we have. Those types of things can happen from time to time. So it's really useful and we use the dashboard for that.

Enrollment is another area where we have to interpret data. We want to see where our enrollment for the coming year is, how we're stacking up relative to other schools. That's a big help too. If we're sort of an outlier at the data point [compared to other schools], then we can start to investigate what we're doing internally that's leading to that outlier status, be it good or bad. But if we notice maybe there's some negative thing happening and we look across the system, and we see that there are 30 other schools experiencing similar problems, then maybe it's a larger systemic problem than just something internally. So I think the dashboard really helps us identify those things. We can use that data to identify need areas, and areas of strength.

Ways a Data Dashboard System Supports Data-Informed Decision-Making

I continued my conversation with Mr. Rodriguez by asking ways in which a data dashboard system supports data-informed decision making. He felt it was especially helpful for simplifying data when talking with teachers. He said:

Typically, most teachers aren't comfortable with data, so what the dashboard does

> is really give us an easily delivered message, as far as data goes. There are a lot of options for graphs and charts. And a lot of people who aren't naturally inclined to look at numbers and data can get a visual representation of trends and that helps us soften the shock that a lot of people experience when they see numbers. So we share with teachers how that data information can help them drive instruction. But as far as parents go, typically our teachers are sharing that information. So we share with the teachers, and the teachers will go share with parents what kind of outcomes they're seeing from their students and what might be driving those outcomes. And also how to adjust in order to address the students' needs based on the data we're seeing.

Monitoring the progress. I asked Mr. Rodriguez how he uses data to identify persistent problems and monitor progress of a school over time. He felt it was specifically useful for monitoring attendance. He replied:

> There are a lot of different areas on the dashboard and attendance is one that we've used to identify attendance issues. If we're seeing particular students with attendance issues, we review that. It first comes to us through the dashboard, and then we have discipline, that of course, is very similar, so the system allows us to easily flag and identify students who need intervention in terms of behavior. Enrollment is huge for us. We've actually struggled quite a bit with enrollment all year, and the dashboard allows us to closely monitor the changes that happen. I don't think most public schools experience that, at least schools with a similar enrollment. Most ISDs don't experience that, so I think the student body doesn't fluctuate as much. Having access to this fluctuation, the regular fluctuation we

> experience is good because the dashboard enables us to do a better job of maintaining our enrollment there.

Accessing the data. I asked Mr. Rodriguez which school data parents have ready to access to. He said he knows parents regularly access the dashboard. Specifically, he said:

> Honestly, I don't know how directly the dashboard ties into the parent portal, but I know they use it regularly because our database ties into the application. They can set up push notifications. My experience at my previous school was really different than here at this campus. At my previous campus, we didn't have much parent interaction; it was a struggle to get parents to the door. Here, we see a lot of parents using the application, and as soon as they get a push that says, "Your student made below a 70" they become involved quite a bit, so they can get access to quick information about student scores, attendance, homework assignments, and things like that. But I really don't know how much of that ties into the dashboard. I think it's mostly focused on the assessment piece, and parents are very aware of assessments, and they're really focused on how well their students are performing relative to others. But I'm not sure what the parent experience is on the dashboard.

Recommendations for System Improvements

In closing my conversation with Mr. Rodriguez, I asked him if he had any recommendations regarding system improvements. He would like to see it used to correlate assessment data. He suggested the following:

There's one piece, and I don't know where the focus should be, but I know we have a lot of smart people working on the dashboard, and one of the things I've always wanted to see would be if we took all the tests, the assessments that we give, and drew a correlation between the various assessments we give and the outcome that the students are going to have on the actual STAAR test or SAT. I would like to see when you take MAP at the beginning of the year I would like to see a range. So, if a student got a score in the 130 range that means they're likely to score in between this and that on the actual STAAR. And then the next time we get a data point, if we could draw a correlation between their MAP score and their first mock STAAR, we could start to narrow that range down over time so we could predict and project where students are with greater accuracy every time we get a new data point. Just a more direct correlation between each assessment and what the outcomes for the kids are going to be.

One of the other things I would like to do is look at how to get the data we're getting from various places into the classroom. So, we'd have all the data present, and it would be a matter of identifying, let's say you have a range, so students that score between 60% and 70% would be grouped together, and 70% and 80% and so on. So if we could get those students to automatically be generated into intervention groups based on whatever criteria the academics department sees fit, then that would [help]. If it went to the teacher that way, they wouldn't have to worry about reviewing that data individually. We could almost have a generated form that would say, "These are the kids you need to see on this day, and these are the topics that you need to cover." And then you could even

> have it refer to the specific information that the academics department is asking us to use for those objectives. It would streamline the process. Right now, we have, I imagine, dozens of teachers, and they're all doing basically the same thing individually; there's not one big organized effort to provide those resources, so it just seems like a massive use of man hours where it could be streamlined a bit.
>
> Data Dashboard would be good if we could tie into the evaluation feed, you could track numbers of walk-ins and things like that, and where you are relative to your goals or full observations. It would also be really great if we could tie in scores, so every time we do a new evaluation, if we score a two the first time, and a three the next time, then it would average out. We could kind of get a better balance of all those in aggregate. That would be great.

Cost information. I asked Mr. Rodriguez how a data dashboard could be used to support evaluation and supply cost information. He had several ideas about how it could be used. He said:

> Regarding cost information, that would be really great. A top vendors list, top expenditures list, maybe even the most frequently used budget codes. What your status is, that would be fantastic.

Sharing the information. Some of the data in data dashboard systems can provide valuable information to the public. I asked Mr. Rodriguez if a data dashboard could be used to share information with the community outside the school. He thinks transparency is important for the school and explained it this way:

> I think we should be as transparent as possible and share what our progress is. If our students are performing well, I think we should share that. If we are a little

disappointed, I think we share that and ask for some assistance from the public. I think they're all more than willing, I think it's just really a matter of helping and we could use the data dashboard to identify our needs and request additional support from the public.

Chapter XII

Data is What Runs Our Campus

Ms. Hudson has been in education for eight years and currently serves as the principal for a charter school system in Texas. She describes herself as someone who likes educating the kids and seeing that "Wow" moment in their eyes when they are learning. It is her goal to find ways to improve academics at her school and find ways to make her campus stronger and better than the other schools in the surrounding area. Her educational philosophy is that teamwork is best, and that team should include the entire campus.

Benefits to Implementing a Data Dashboard System

I began my conversation with Ms. Hudson by asking about the perceived benefits of implementing a data dashboard system for administrators, teachers, parents, and students. In her opinion, a key benefit to the dashboard is having all the information in one place. She said:

> It will be easier for us to be able to track [information] and when we need something to be done on campus, to be able to put that into a dashboard so that it will go directly to the right people instead of our having to send out numerous e-mails to various people throughout the day. With one system it will go to those people directly.

Internal needs. Having access to a data dashboard impacts schools. I asked Ms. Hudson about the internal needs for creating a data dashboard. She feels the data dashboard would improve the education students receive. She explained as follows:

> I would need an IT person. I do not have that now, and we would need training just to make sure we knew how to input into it, and put the needs in there, and how to respond, and how to track back and forth. So, a good training would help. Having access to the dashboard would give us, as administrators, more time to be in the classrooms instead of doing the busy work and the paperwork, and that's what we need to be doing. We need to be in the classroom, helping the teachers, making sure the students are receiving the best education they can receive.

External needs. A data dashboard system can also assist in meeting the needs of external customers, such as parents. Ms. Hudson believes the dashboard will allow the school to reach more of the community. She said:

> There is a need. I think we could hit more stakeholders or more community people with the dashboard if they had access to it to see what is going on at the campus. Externally, we would need a company to be able to create a dashboard, or somebody over there in Cosmos that could create a dashboard for us, but for the community and for parents and external things, we would want them to understand how the dashboard was run.

Interpreting the data. Being able to interpret data is an important aspect to implementing a useful system. I asked Ms. Hudson how she uses the data when she has access to it. She responded as follows:

> Data is what runs our campus. I pull up the data all the time looking at benchmark data, looking at attendance data, looking at mark star data. The data kind of runs my weekly meetings with all the people I meet with on a weekly basis. The

> teachers know the data; the front office knows the data that they need to know about. So, it's how our school runs. It's based on the data that we receive.

Ways a Data Dashboard System Supports Data-Informed Decision-Making

I continued my conversation with Ms. Hudson by asking her how she uses a data dashboard system, and whether she could use it in conversations with parents and staff. She feels the data helps her school see how it compares to other schools so they can make improvements. She said:

> As a school system, and even as a district, it would give us information to see how many students have DTS. Is our discipline system working? It would also show us, as a system attendant, why some campuses have higher attendance than others and what they're doing. And on the academic side, in addition to looking at the benchmarks for the mark stars, we could communicate with other campuses and see what they do differently, and how they achieve their scores.
>
> With staff, I do meetings with them about data bi-weekly and weekly during our PLC time. We are always talking about data. With parents, we use the data for the academic side. And we use the data for attendance as well when we meet with them.

Monitoring the progress. I asked Ms. Hudson how she uses data to identify persistent problems and monitor progress of the school over time. She said she uses the data to spot trends and make improvements. She explained:

> We look at the trends. If we see that there is one, such as a particular student or a particular subject area that is constantly popping up on the data as not being successful or causing problems in the classroom, we use that data to see what's

> going on. What can we do differently? How can we change either the behavior of the child, or how can we academically help the teacher, support the teacher or the students in the classrooms?

Accessing the data. Data dashboards offer several options for accessing data. When I asked Ms. Hudson which school data parents have ready access to, she responded as follows:

> The data they have access to right now is the student's benchmark, the student's mark stars, and the AR test. I would like to see parents be able to see the data for other computer programs that we have such as ALEKS, Study Island, I-station, and Accelerated Reader. For that type of data, I'd also like them to be able to see the student's attendance data throughout the school year.

Recommendations for System Improvements

In closing my conversation with Ms. Hudson, I asked if she had any recommendations regarding system improvements. There is information she would like to see added to the dashboard. She suggested the following:

> With the dashboard we currently have, I'd like to add on there the computer usage so we would not have to go into every single computer program that we have to look up the usage on a weekly or bi-weekly basis. I'd like to be able to see that. As for the rest, we've got the benchmarks in there, now we have the mark stars in there, that's probably the only other thing that I would really like to see on the dashboard at this point. As for evaluations, we could use it to monitor the teachers in how they're doing in the classroom such as behavior-wise and academic-wise.

Cost information. I asked Ms. Hudson how a data dashboard could be used to support evaluation and provide cost information. She feels that having the data in one place is very useful. She said:

> If we could tie JR3 financial software with that, as well, that would be good, too, instead of having, again, to go to a different site to prove our POs, to monitor our budget, etcetera. If we could have everything in one area, that would be greatly appreciated and helpful.

Sharing the information. Some of the data in data dashboard systems can provide valuable information to the public. I asked Ms. Hudson if a data dashboard could be used to share information outside of the school. She said:

> Academically, we can share the data. We can also share what type of computer programs we have for the kids. And we can share, overall at the campus, discipline-wise as well how discipline is done on this campus or how many GPS points versus TRS points could be shared. But we have to keep some things private and not disclose kids' names, for example, so the public would just have an overall, general look at it.

Chapter XIII

Data Dashboards Provide Several Options for Accessing the Data

Mr. Sanchez has been in education for 11 years and has been the assistant principal for academics at a charter school system in Texas since 2013. In his present role, he is also the testing coordinator and the special education coordinator. Mr. Sanchez describes himself as someone who cares about education and looks out for the educational needs of the less privileged. He explained that he tries to promote the idea of social justice to those students he feels are forgotten. He said, "You have minority students or disabled students, and that's a big interest of mine in education."

Benefits to Implementing a Data Dashboard System

I began my conversation with Mr. Sanchez by asking about the perceived benefits of implementing a data dashboard system for administrators, teachers, parents, and students. Mr. Sanchez believes a data dashboard will benefit everyone, including faculty, staff, students, and their parents. He explained it this way:

> A system like that, like a database system, will have information easily available so that the communications between X school and the parents can be easily obtained. However, in order for that to happen, the school has to make sure that the system is operating properly, and that it's user-friendly. And that someone is responsible for maintaining it at all times. That way, parents and the staff won't get discouraged if it doesn't do what it was promised to do.

Internal needs. Having access to a data dashboard impacts schools. I asked Mr. Sanchez what he thought the internal needs were for creating a data dashboard. He responded:

> Strong IT support that will make sure it's running smoothly, without much maintenance that needs to be done on a constant basis, is important. Information needs to be readily available, and the less involvement necessary, the better. That would really help the education part because it would eliminate a lot of these logistics questions parents have, or even teachers, and a lot of that could be addressed through the database. So, that will mean less time will be spent trying to run a school having to worry about the information that could be done in the database.

External needs. A data dashboard system can also assist in meeting the needs of external customers, such as parents. Mr. Sanchez is aware that parents access data now, but he is not aware of any pressing need for additional data sources. He said:

> Just speaking about this campus, I think we have a database that provides staff and parents the information needed, and I personally haven't heard parents or staff express any concern about information that could easily be obtained through a dashboard, or whatever. I personally have not heard of anything of that sort. I know, as a matter of fact, when I talk to parents on a repeated basis, when we're talking about grades or anything else, they always mention, "Well, yesterday I was on my son's database and so I saw this, I saw that." And so parents use it, what we have here, parents have access to it and they actually use it to make decisions. So personally, from what I have heard and experienced here at this campus, I don't see a big concern for improving, or even having these separate dashboards. Improving smaller things might be of benefit, but personally, I haven't heard anything.

Interpreting the data. When I asked Mr. Sanchez how he interprets the data he has access to, he explained that he likes to use a combination of hard and soft data. He described his feelings this way:

> When it comes to data, there's a large sample of data. I mean for instance, me in my role, I use academic data, I use data from benchmark assessments, from STAR, from MAPS and so on. I use that data. I use also data from the teachers, and I use a lot of soft data. Personally, I use a lot of more soft data than I do hard data. Soft data is the communications that I have with the teachers when I just sit down and talk to them and see how they're doing. I do a lot of that. I think to me personally, that type of data brings a lot more meaning to the numbers and the figures than the hard data brings. So, whenever I have the hard data and numbers, I look at that, but that only gives me numbers, that doesn't really give me any meaning. So that's where I try to go and actually talk to people, to try to bring some reason to the other type of the data. And then from there, once we have all the different types of data, the soft and the hard data, then that's when we need to create plans to improve the specific areas that need to be addressed.

Ways a Data Dashboard System Supports Data-Informed Decision-Making

I continued my conversation with Mr. Sanchez by asking ways in which a data dashboard system supports data-informed decision making, and how he could use it in conversations with parents and staff. Mr. Sanchez feels the real benefit comes from all the information in one place. He said:

> When I have the data that I access from our database, all it really does for me is create just the surface of the issue that we're trying to look at. For instance, if I'm

trying to create an intervention, or I'm trying to create whatever, I can run data from the database and have that, but all it gives me is numbers; it gives me particular low scores here and high scores there, and so forth. But data in itself to me is not sufficient to make the decision that will improve the students. So, that's why in using any dashboard to accumulate data, to me that's where people do great in having different options where you can easily gather that data. In gathering that data, I think a dashboard will be great because it would be easily accessible. And you can go locate it from one location, as opposed to trying to hunt it down from three different, or four different locations. And [the dashboard] can be used as the base to acquire whatever data you need. And then from there, once I have that, then the data is no longer needed, or the dashboard is no longer needed, because that's when we kind of make more sense of the data in a more personal way, where communication happens.

So in my role as academic, when I use data for academic purposes, and I'm speaking to parents I kind of just bring in the facts about what the data is showing, and really, they can see in a very simple way what it's trying to say. And I can review with the parents the STAR scores, or whatever. But then I kind of move away from the data, and again, kind of make sense of the data in a more personal way. When we talk about data, it can be very dangerous if people don't use it correctly. And so I try to give data a place, but I don't emphasize the data so much, because the data misses a lot of factors that cannot be generated in a report. So whenever I review data with teachers, I kind of just give them, "This is what the data is saying according to just this piece of data, but let's now make more

> sense of it." And so we move away from that and try to fill in the gaps in the data that, unfortunately, reports don't give out. Like, for instance, factors that can contribute to students not doing well in class, "What's happening at home? Have we talked to parents? How many DPS points does the student have? Do we see any other behavior problems? Or are things bad at home? Are these kids having to work after school, and that's why things are not getting done?" Things that, unfortunately, cannot be measured by the data we have.

Monitoring the progress. Data dashboard systems can also assist in monitoring progress. I asked Mr. Sanchez how he uses data to identify persistent problems and monitor progress of the school over time. He replied:

> Just speaking about persistent academic problems, because it's what I do, I personally look at the different assessments we have. Whether there is a trend in these students, and then compare that with the data that I received when I talked to teachers. I compare that data to the soft data I receive when I sit down and actually talk to the teachers and say, "Well according to STAR results and according to the factors, or the district assessment, this student is not doing well. But in your class, this student is doing very well. What can we decide? What can we come up with that perhaps is the problem?" So that's kind of how I use the data.

Accessing the data. Data dashboards provide several options for accessing the data. When I asked Mr. Sanchez which school data parents have ready to access to, he responded the following:

> The data that parents can access right now are student grades, and DPS, PRS

> points. They have access to student's assessment scores, again through us. But academic data, I think that's about it. And I'm not too sure what other data parents would need that would really be meaningful information for them. Personally, I don't know if other data would be needed.

Recommendations for System Improvements

In closing my conversation with Mr. Sanchez, I asked him if he had any recommendations regarding system improvements. Because he does not rely heavily on hard data, he did not have any specific suggestions. He had this to say:

> I don't think that it needs anything more, I think what is accessible right now is sufficient to make the decisions, or even more than enough to make the decisions that we need to make when it comes to educating our students. I'm never a proponent of data that unfortunately sheds light on the good students, and sheds light on the bad students. I don't think it's a good practice to have data that will automatically already separate the good and the bad students, because in reality, most data sheds light on the students who already are going to be good, regardless. And it sheds light on the bad students, regardless of what it is. So, that's why data needs to be very carefully analyzed, because for the most part, it always does what it's intended to, to show those students who are doing very good, and to show the students who are doing very bad. I mean, that's the history of data, it's academic data. So that's why I don't think we need any more data. I think we have more than enough. I think what we need more is the soft data, I think we need to be out more and talk to the teachers and have more of that kind

> of communication that is more meaningful than just the qualitative data. Again, that's my very personal opinion.
>
> And going back to the earlier questions, we can easily have a dashboard that will record when a teacher doesn't do the district assessments on time, or when a teacher doesn't do the math tests. So we can have a dashboard that will record all these times, or we can have a dashboard that will record how many walk-throughs the administrator does a week, so forth and so on. So that can be a form of maintaining an evaluation on these people, but again, we go back to the questions of what are we really trying to do? Are we trying to control and intimidate? Or are we trying to really progress and have the human element behind the education, because on an administrator level, if you're required to do five walk-throughs a week, and there's a dashboard that reports you do only two a week, when somebody just looks at that, they're going to assume, "Well, they're not doing their job." Or something like that.

Cost information. In wrapping up our conversation, I asked Mr. Sanchez how a data dashboard could be used to support evaluation and provide cost information. He replied as follows:

> For the financial aspect of a data dashboard, the only way that I can think of, is just inventory. That's really the only way, and maybe also if teachers want to put in to buy a bookshelf, or I don't know, books or whatever, there can be a place where teachers can put that suggestion in, and then the finance people can go through that list and determine what the purpose of that bookshelf is. But that's probably the only thing I can come up with.

Sharing the information. Some of the data in data dashboard systems can provide valuable information to the public. I asked Mr. Sanchez if a data dashboard could be used to share information with the community outside of school. He does not see any real need to share information beyond the parents. He explained his views this way:

> With the general public, I would think that it can be used in a way to promote, or basically to show maybe attendance rates. And I guess the question would be, "Why would we want to?" I mean for grade purposes, I mean for parents, I think parents would really benefit from grades and the behavior things, and that's already set for the parents who are doing that. But aside from that, I don't know if a dashboard necessarily needs to be shared with the public, because a lot of that is internal information that meets one goal and hopefully that goal is to promote a healthy and safe learning environment. But a lot of that is already happening right now, because parents already have access to that.

Chapter XIV

The Data Dashboards Are Helping Us Get the Data Quickly

Mr. Johnson has been in education for 11 years, and currently serves as the principal for a charter school system in Texas. Mr. Johnson is a big believer in teamwork and has experience in all aspects of the school because he has dealt with teachers, parents, and students. He believes there has to be team effort for a school to reach its goals for excellence. His philosophy is that stakeholders, students, parents, teachers, and other community members must work together to achieve campus goals.

Benefits to Implementing a Data Dashboard System

At the beginning of our conversation, Mr. Johnson told me that data dashboards were a new and growing area for his schools. I asked him what he felt the benefits of a data dashboard might be. He said:

> We were introduced to data dashboards in the last couple of years and we are still learning, but I see great benefits in using a data dashboard because in general, I can say it gives school administrators, or the users of the dashboard, live data about all aspects of the school. And it allows us to see what direction our students or our school is going. You can see the numbers; you can compare the numbers. Starting with attendance, schools are getting their state funding from attendance and it's one of our goals to keep our attendance at a certain level. Our students will be here attending our school and the school funding will begin coming properly.
>
> In the past, it was difficult to keep track of the school attendance on a daily basis, even for a day or certain hours, certain periods. Now in the data

> dashboard, we are able to see every day how students are doing in terms of school attendance or in terms of our DPS numbers and students' credits, gaining credits. Just as an example, data dashboards are helping us with attendance, keeping track of our students' college revenues, setting goals, keeping up with our accountability goals, enrollments, operational needs, discipline, lunch, and so many other things. So I would say in general, yes, having access to all this data readily and using it as we need to set our goals and just communicate with our students, parents, and teachers is beneficial.

Internal needs. I asked Mr. Johnson about the internal needs for creating a data dashboard. He said that their needs are derived from their goals and went on to say:

> Schooling is a complicated and complex job, or profession, or thing to do. We are dealing with students and parents and teachers – humans – and it's very complicated, but we have certain goals set by our school administration, our central office, or by our state for accountability, and we have federal rules and implementations in place. So considering all those rules, requirements, accountability requirements, school goals, all those are trackable or doable by keeping track of the data in different ways. For a state it could be accountability for state testing, for schools, having their students go to good colleges, attend top-tier colleges. But how do you get them ready for college? You have to keep track of their college readiness.
>
> For operational or federal requirements, we have lunch programs, our attendance, and grants, and every single one requires keeping track of data. So the dashboard is a huge need for us because it gives us access to all those data in one

> place. It's really important to have all of it in one place because we deal with so many other things in the school and, if you have your data in many different platforms, it is difficult to gather and plan and work toward your goals. If you have all of it in one place, easily accessible, it makes it easy for us and it answers all our questions about keeping up with our goals and requirements. So I would say data dashboards are needed to keep up with the state, federal, district, and school goals.

External needs. About the external needs for creating a data dashboard, Mr. Johnson emphasized parents' needs, but added that state and federal agencies and the community benefit as well. He described parents as main stakeholders and said:

> They send their kids to our school. They can get a good education, they can be ready for college, and they can attend good colleges. They want to keep track of their child's education, but they're not able to come to school every day, so the solution is to provide them easier access to their child's progress. The data dashboards are allowing them to see their kids' progress in all different aspects of schooling. You know, attendance, grades, college readiness, state tests, college readiness tests. Parental communication is one of the main external needs for creating data dashboards.
>
> On top of parental communication, I think of the state and federal requirements. We are often asked to submit data for certain state or federal requirements, or grants, or other things, and we have to provide that data. So the data dashboards are helping us get that data quickly, make our submissions, and provide data to those external institutions.

> Another reason would be increasing our community engagement. We are using our data to increase our community engagement. I've been thinking about this, actually. The current data dashboard we have doesn't do much of that, but I think it's something that can be added. We use a lot of community outreach programs or platforms, starting with the school website, Facebook, Twitter, and many other things. We inform people. We take some data from our dashboards, of course. We are using it, but I'm referring to having our data dashboards communicate with our outreach platforms like Facebook, Twitter and school websites. But we are using the data from the dashboard to inform our community, as needed, on a daily basis.

Interpreting the data. When I asked Mr. Johnson how he interprets the data, he said he could probably give me several examples. He specifically talked about accountability goals and how data can be used for comparisons from year to year. He explained it this way:

> Because we are a public school, every year we are expected to reach certain goals by our state and by the federal government. We have accountability goals; we have to meet our standards or go beyond those. We have to consider all those goals set by our state or federal government and internally we plan to achieve those. Every year we set some goals for our teachers, for our students, and when we set our goals we use the data. We are not just looking at one year's data; we are comparing previous years. Two years ago how did we do? How did last year compare to the previous year? What is our goal for the following school year? We

> use our data to set our accountability goals to reach our state or federal accountability.

Ways a Data Dashboard System Supports Data-Informed Decision-Making

I continued the conversation with Mr. Johnson by asking ways in which a data dashboard system supports data-informed decision making, and how he could use it in conversation with parents and staff. He told me that almost all their decisions are based on the data in the data dashboards. He further said:

> I think those are the main things I can give as an example of using the data dashboards and how it is informing our decision making. We communicate with our staff all the time and talk about setting goals. Within our school system, we work toward certain goals set by our team. We set the goals together. And even starting with the setting of goals, we use the data dashboard and the data in the data dashboard for each one of our teachers. I check their progress from previous years to see how they did in terms of accountability or the assessment and the set goals for next year.
>
> And when I'm setting the goals, when we are talking or keeping track of the progress toward those goals, we talk with our teachers. We meet with our teachers one-on-one and look at the data from the dashboard or the different platforms we use. We communicate.
>
> We talk about how we are doing in terms of our goals. Say the teacher's goal is 80% and the recent benchmark, or the recent test, shows the teacher is a little behind the goal, say 70% compared to 80%, then we start digging the data to see the possible reasons. How did we do compared to last year, for example? So

we go to that dashboard or the data platform and see how the data compared to last year, and if there are differences, we start questioning ourselves, the team members, about why. What changed from last year to this year? We start looking at our curriculum. We start looking at our assessment tools. Then we start looking at the instruction in the classroom with walk-throughs or observations. Then we get different data and compare. We identify all the possible reasons, consider them, initiate interventions and move on.

With parents, we use the data dashboard in streaming several areas, and we should communicate with them on a regular basis. Our dashboard for our students' database allows parents to see their student's progress in terms of attendance, grades, college readiness, testing or almost any other thing. Sometimes parents don't have access to those dashboards. We give them access, but they may not have a computer at home or other device to check, so they might not be able to see. In that case, our counselors meet with our parents to communicate their student's progress, and when they meet with them the data is in our dashboard, and can show them how their students are doing compared to their goals and talk about what they need to do. We discuss all that with parents.

Another way that we use the data is attendance. Students are expected to come to school every day, but sometimes because of sicknesses or other reasons students might not be able to come to school. They miss the instruction and we cannot afford that. So we meet with our parents and use our data dashboard to show them how their child's progress is based on attendance. We can show how the child was doing before the absences, and how the child is doing after he or she

> is absent from school; how their performance changes. We can show those outcomes to parents and expose what happens when a child misses instruction; how they can't progress. So we can communicate with our parents using the data in the data dashboards and make them understand.

Monitoring progress. I asked Mr. Johnson how he uses data to identify persistent problems and monitor progress of a school over time. He told me they use it to track school attendance and added:

> When we do our campus improvement planning, we look at our attendance data for each school year. In the school year we look at each FSP period, and every six weeks we submit our data. In order to improve, we compare our data for previous years to this year and we set some goals. So there's another use for the data from the dashboard. The dashboard allows us to see how we did in terms of attendance. How we did last year, and how we did the previous year. And what our goal for this year is. We keep track of our progress after setting the goals on the data dashboard. I can simply see it. Every day I can go there and see what my school's attendance percentage is, how it changes from day to day, and where am I in terms of reaching my goals: My goal is 97 or 98 and if I'm doing 96.5 then I can communicate with my team and say, "Hey, we are behind. Let's come up with some intervention, some ideas to reach our goal."
>
> On a daily basis we use our dashboards or data platforms to keep track of our students' progress in terms of discipline and behavior. The data dashboards are simply allowing us to see all of our students' progress in terms of behavior from day to day, and we can, based on that progress, see if they're regressing or if

> their behavior is not going well. We can see it with the numbers in our systems, and we can take interventions, we can plan dual assemblies, and plan one-on-one parental meetings.

Accessing the data. When I asked Mr. Johnson which school data parents have ready access to, he gave me several examples. Grades were the first things parents want to see, but there is other information as well. He explained:

> Parents want to see their child's progress in the school for each course or each subject they're in, like science, math, physical education, all of them. The students' grades are readily accessible from the school database. They can keep track of their students' attendance and see whether they were at the school or if they missed any of their class periods, and they can take intervention and communicate. So they have access to attendance. They have access to the student's state testing or school lights benchmarks or mark testing, basically the test results. They can easily see the student's test results. They can keep track of their homework on the student database, and I think those are the main things. And discipline too, they can keep track of their student's discipline or behavior progress. Our student database will allow them to see it on a daily basis.

Recommendations for System Improvements

I asked Mr. Johnson if he had any recommendations regarding system improvements. I also asked him whether he thought a data dashboard could be used to support evaluation. He had the following suggestions:

> We have so many other data platforms that we use, but the dashboard makes it easy because we can see everything in one place. We are using a different

platform for evaluation, and I'm not sure how easy or difficult it would be to get that data on the dashboard, but if it's possible, it would be so great to see our teachers' progress there, too, because we can see the student progress, we can see almost every test or every data or every evaluation result for our students, but there is almost nothing about the teacher progress.

Another dashboard has some college-readiness portions. It allows us to see the students' college-readiness data - basically college-readiness tests, either the PSAT, SAT, ACT, or other tests. But it doesn't allow us to see the whole college-readiness aspect in one place.

I'd rather have a dashboard for students that shows all of their college-readiness goals. College-readiness is not just about the test. How are they doing in terms of their volunteer hours? It's a requirement that our students do 100 hours of volunteer work, but there's almost no place that we keep track of that on their dashboard. We can see it in the database, but I have to go and check every student one by one. If we had it in the dashboard as a graph, something that's easily understandable, that would be really good.

And just like that, their credits. Yes, we see the speed test, but what are you going to do if you have your students not graduating? First, you have to keep track of their credits, the required credits or their recoveries. It's a big hassle for us. In the field we use some extra documents, Google documents, to keep track of our students' recovery or credit needs and volunteer hours. So, I was thinking, if we could have the graduation and college-readiness all in one place for students with some graphs that would be really nice.

Cost information. I asked Mr. Johnson whether he felt a data dashboard would be useful in monitoring costs. He felt having that information would allow him to make better decisions. He said:

> I would like to see school expenses on a data dashboard. It would be very nice to have that feature. That way, I could track how much money is spent for each item. It would help me make better decisions for remaining amounts, especially if this information were easily accessible on the dashboard.

Sharing the information. Some of the data in data dashboard systems can provide valuable information to the public. I asked Mr. Johnson if a data dashboard could be used to share information and how he would use it. He said:

> In general, most of our success and testing data are published through our school website. I believe it would be great if the data dashboard pushed some data to our school website automatically for public information, such as attendance rate, college acceptance rate, etc. This would be good for transparency to the public.

Chapter XV

Narrative Analysis

The purpose of this qualitative phenomenological narrative study was to explore the perceptions of administrators regarding the implementation of a data dashboard system in one Texas Charter School System. This study asked 10 administrators who were currently working as campus principals, assistant principals, and area superintendents to share their experiences with the implementation of a data dashboard in their schools. This chapter analyzes the responses of the 10 people interviewed and presents the emergent themes from those interviews. Findings are reported by the following research questions:

1. What do school system administrators perceive as benefits to implementing a data dashboard system for administrators, teachers, parents, and students?
2. What do school system administrators perceive as ways that a data dashboard system supports data-informed decision-making?
3. What topics should the dashboard system encompass that are currently not being addressed?

Research Question One

Research Question One asked school administrators what they perceived as benefits in implementing a data dashboard system for administrators, teachers, parents, and students. Emergent themes included existing access to data, reviewing the school data, community needs, and advantages of a data dashboard.

Existing access to data. All the participants mentioned that they now use various software to collect data for different purposes. In addition, they have to access these different platforms through different accounts. Four out of 10 participants talked about how it was difficult to access data on so many different platforms. Mr. Sean expressed that his staff members always need access to data to understand how students are doing, and what they could do to improve the students' learning because there is so much data. Both Ms. Bee and Mr. Johnson explained that gathering data from different platforms and creating spreadsheets for each data set made it difficult to plan and work toward their goals. In addition, Ms. Bee felt that without a dashboard, there was no way to access something in a visual way that explains the data very quickly.

Reviewing the school data. The benefits of being able to review school data on a data dashboard emerged as a common theme among the participants. All the participants agreed that school administrators need to review the data and look for red flags. Mr. Scott told me that he always makes a comparison from year to year to predict students' performance at the end of the year. In addition to this, Ms. Thomas said:

> When we have all of our data together, we see whether there is a trend. For example, if we are having trouble in third grade, [we want to know if] it is just third grade math, or is it third grade math and reading? Then is it the teachers or the students? So we look at the data to determine what our issues are.

Ms. Bee agreed with what Ms. Thomas said and added that her team noticed that by using a data dashboard the team could identify that some students were not progressing and they needed to change their intervention strategies.

Community needs. Another emerging theme of the participants with respect to implementing a data dashboard was community needs. Nine of the 10 participants mentioned that everyone is looking for reliable data with easy access to understand how the schools are doing. Mr. Sean expressed, "Parents, community members, and stakeholders want to know what is going on in the schools." Another participant, Mr. Scott, a principal said, "Our parents are asking for tools from schools and from us to see their students' progress." In addition, Ms. Hunter stated that with a data dashboard, parents and the community can see the rankings of the school with testing, and they can look at how the teachers are performing.

Advantages of a data dashboard. Six out of the 10 participants stated that the data dashboard provides quick access to data coming from different platforms. Mr. Scott and Ms. Hunter mentioned that the data dashboard gives a lot of information in one place very quickly. Mr. Rodriguez pointed out that the dashboard allowed his school to centralize the information. He said, "Having all the data gathered together gives us a broader, more encompassing perspective than they have when they see it in individual pieces." Mr. Johnson added that having readily accessible data meant they could use it more effectively to set goals and communicate with students, parents, and teachers.

Research Question Two

Research Question Two investigated whether school administrators felt there were ways a data dashboard system would support data-informed decision-making. Emergent themes included monitoring progress, supporting data-informed decision-making, communication with stakeholders, and existing access for parents.

Monitoring progress. The first emerging theme that seven out of 10 participants mentioned was monitoring students' progress on the data dashboard. Mr. White stated:

> We use the data dashboard all the time. For example, when we meet with our principals, we go to the data dashboard and look at the enrollment numbers. "What is the enrollment number today? What was our goal? And what is the difference now? What should we do if enrollment numbers are below our targets?" So, the data dashboard helps us to have access to crucial data immediately, and we can start a conversation with the principals.

Another participant, Ms. Thomas, an assistant principal said, "We use the data dashboard to monitor our ESL and our special education students. Are they growing academically? Or are they staying the same? And then we identify how can we help them to increase their scores." Ms. Hunter believed that the teacher is the key in instruction and said:

> When I start looking at data that shows a teacher is constantly struggling, I try to figure out why? Is it because they don't understand how to teach the content that they're teaching, or do they not have the right resources? So I look at the data and I determine how many students passed, how many students failed, and what was it they failed? How can I help this teacher figure that out? Once I help figure that out with the teacher, and I get them signed up for training, or get them hooked up with a mentor that can guide them with some strategies.

Supporting data-informed decision-making. Another emerging theme was supporting data-informed decision-making. Almost all the participants mentioned that the data dashboard helps them make better decisions. Mr. Sanchez, an assistant principal

stated, “Whenever I review data with teachers, I kind of just give them, “This is what the data is saying according to just this piece of data, but let's now make more sense of it.” Another participant, Mr. Johnson said:

> We communicate with our staff all the time and talk about setting goals. Within our school system, we work toward certain goals set by our team. We set the goals together. And even starting with the setting of goals, we use the data dashboard and the data in the data dashboard for each one of our teachers.

In addition, Mr. Sean said, “At one of the schools I supervised previously, we had reading specialists and math specialists guiding teachers to make better decisions. But, without a data dashboard, this is a time-consuming task for everyone.”

Communication with stakeholders. Eight out of 10 participants emphasized that data is crucial for all stakeholders including students, parents, and community members. Mr. White stated, “We have very positive experiences with data dashboard, and we have been using this for all of our parent conversations.” Ms. Thomas agreed with Mr. White and noted:

> Well, with our parents, when we have our annual CIP meetings and our title one meetings, we review stuff like: How did we do on testing? How are we doing on attendance, our growth? So we use all this data that we have to relate to our parents so that they see... that we are a strong school and how we help make students stronger here on campus.

Mr. Johnson also expressed the importance of data dashboard in communicating with stakeholders. He said:

> We are using our data to increase our community engagement. We use a lot of community outreach programs or platforms, starting with the school website, Facebook, Twitter, and many other things. We inform people. We are using the data from the dashboard to inform our community, as needed, on a daily basis.

Existing access for parents. The last emerging theme was the ways that a data dashboard gives access to parents. Out of 10 participants, nine mentioned several options for parents accessing the school data. Mr. Sean, a school principal said, "Our parents have access to a mobile application where they can see their child's daily grades. They can email teachers from the application. They can see attendance records, discipline records." Another participant, Mr. White stated, "Our parents do have access to data they need, and this helps us have meaningful conversations with parents." Moreover, Ms. Hunter mentioned that the data dashboard gives access to parents, and she noted, "By accessing the data, the parents think, 'Where does my child need to be, and what do I need to do as a parent to offer more support and help for my child? Or is it the school? And is that a concern I can bring up to the school?' "

Research Question Three

Research Question Three asked school administrators what kinds of data they would like to see included on the data dashboard that are not currently being addressed. Emergent themes included supporting evaluation, cost information, sharing the information, and additional features needed.

Supporting evaluation. Six out of 10 participants mentioned that a data dashboard could be used to support evaluation. Mr. Sean stated:

> Evaluating teachers, students, and administrators is very important because it tells him what is working and what is not. If data is not readily available, it does not get used. Without that data, it is difficult to determine and compare progress. The availability of a data dashboard would facilitate evaluation discussions. A data dashboard allows the administrators to talk to teachers about how they are performing and where they need to improve.

Another participant, Mr. White, an area superintendent noted, "Schools use different software for different areas. For example, for teacher evaluation, we are using company based software to enter all the data. It can somehow be transferred to our current data dashboard system." Mr. Scott agreed with Mr. White and said:

> Right now, we are using different software for curriculum tracking to see where a teacher is as far as instruction and progress. We do not have any curriculum data transferred to the data dashboard. We have to go to a different platform to check it. It would be nice to see this information on the data dashboard because of its easy access. I prefer to have access to all software from one location.

Cost information. Another emergent theme was how a data dashboard could be used to provide cost information. Almost all participants agreed that it would be great if financial expenditures were shown on the data dashboard. Mr. Scott stated that he could track how much funding his school gets from the state, and where the money is being spent. Another participant, Ms. Thomas said, "We could actually see what our budget is for, let's say, our special programs, our materials budget; and we could see what we've spent over the year and what's available for purchasing." Ms. Bee, a campus principal also said:

> Budgeting will be very helpful to have in dashboard. I have been working towards trying to get a way to make our budget a little more understandable to our teachers so they feel as if they know what our campus expenses are, and understand when we have money, what we want to spend it on, the priorities. And involving our teachers, and mostly our department heads and leaders on campus, in making some of those decisions of prioritizing through our CNA and CIP meetings.

Sharing the information. Seven out of 10 participants mentioned that a data dashboard could be used to share information with the public. Mr. Sean stated that the public wonders how much schools do spend, how they do perform academically, and continued by saying, "Everyone wonders about these. Having access to these data publicly somewhere on our website will help them understand our school." In agreeing with Mr. Sean, Mr. White said:

> We are a part of the community. So, they need to know who we are, what we are doing, and what our performance looks like. I can also say that it is very critical for us to be transparent and share the information with the public.

Another participant, Mr. Rodriguez noted:

> I think we should be as transparent as possible and share what our progress is. If our students are performing well, I think we should share that. If we are a little disappointed, I think we share that and ask for some assistance from the public."

Additional features needed. Almost all participants expressed that there could be some additional features needed for the current data dashboard. Mr. Johnson stated that it

would be great, if there could be graduation and college-readiness all in one place for students with some graphs. Ms. Bee said:

> The curriculum pacing guide we have, and which we currently have to log in to a separate database for to pull a report for our teachers, if that were all in one place it would be really helpful. It's a great tool to kind of just check on where things are at. I know that we pull it, and we tell teachers "We're seeing this. We're seeing your lesson plans through here, and talk to us a little bit about your plan to make up this gap," or whatnot. So I think that incorporating that will be super helpful for us.

Another participant, Ms. Thomas, an assistant principal for academics stated that at-risk numbers are shown on the dashboard, and she said, “That is part of our CNA. Since I do both CNA and CIP, that would be really great to have on there... to see our at-risk population on the data dashboard.”

Summary

In Chapter XV, the narrative analysis of the findings was presented. In Chapter XVI, the researcher reports a summary of the study, conclusions, implications, and makes recommendations for further studies.

Chapter XVI

Summary, Conclusions, Implications, and Recommendations

This study explored the perceptions of administrators regarding the implementation of a data dashboard system in one Texas charter school system. This chapter includes the following sections: Summary of the Study, Conclusions, Implications for Practice, Recommendations for Future Research, and brief concluding remarks.

Summary of the Study

Kowalski et al. (2008) stated that school principals have great responsibilities to make data-driven decisions at their schools. Similarly, Tomic and Milic (2013) explained that when data is not well presented, it has no meaning. They further suggested that when data is presented in a meaningful format it can provide actionable insight (knowledge) into how business is doing. Moreover, Blau and Presser (2013) stated that online school information management systems have become an important tool for effectively managing pedagogical information and student data. The purpose of this study was to explore the perceptions of administrators regarding the implementation of a data dashboard system in one Texas charter school system.

Overview of the problem. Bremser and Wagner (2013) expressed that data, by itself, is meaningless. They stated that when data is displayed on a form with graphics and a summary it becomes information that is understandable by everyone. According to West (2012), the dashboard provides all stakeholders with the critical information they need to effectively interpret and act on data in order to keep everyone on track for student, school, and district-wide goals. Similarly, Lange et al. (2012) found that school

leaders and teachers need many different kinds of information in order to interpret and act on data information about student progress including student achievement data, demographic data, school process data, and perception data. Bremser and Wagner (2013) expanded on this by noting that because dashboards provide users with quick and easy access to real-time information, users can more effectively interpret data. As a result, the use of customized user-friendly dashboards has recently increased.

Purpose statement and research questions. The purpose of this qualitative phenomenological narrative study was to explore the perceptions of administrators regarding the implementation of a data dashboard system in one Texas charter school system. For the purpose of this study, a data dashboard system is defined as a tool to share key performance indicators and any other relevant information to all stakeholders in real time and in an easy format (Ballou et al., 2010).

The following three specific questions were used to guide data collection and analysis:

1. What do school system administrators perceive as benefits to implementing a data dashboard system for administrators, teachers, parents, and students?
2. What do school system administrators perceive as ways that a data dashboard system supports data-informed decision-making?
3. What topics should the dashboard system encompass that are currently not being addressed?

Review of the study design. This was a qualitative study using a phenomenological narrative approach in an attempt to explore the perceptions of

administrators regarding the implementation of a data dashboard system in one Texas charter school system. According to Creswell (2013), "Qualitative researchers use an emerging qualitative approach to inquiry, the collection of data in a natural setting sensitive to the people and places under study, and data analysis that is inductive and establishes patterns or themes" (p. 44). Creswell (2013) also indicated that the color of the participants, the characteristics of the researcher, and the detailed analysis of the problem are main components of the ultimate product. In addition, Creswell defined phenomenological study as describing common meanings from lived experiences of individuals around a phenomenon. The stories of participants formed the data of this study and were used to identify themes.

Participants. This study was conducted at different campuses of the largest charter school system in Texas. 10 administrators from these campuses were selected via purposeful and convenient sampling. A purposeful sample was used to provide an in-depth understanding of the phenomenon. Creswell (2013) suggested that convenience sampling be used when the participants were available and willing to participate in the study.

Participants were selected from a pool of possible candidates based on an initial screening process. Specifically, participants met the following criteria:

1. The participant was a charter school administrator serving as a principal, assistant principal, or area superintendent.
2. Each administrator had two or more years of experience in the district and in their current position.

3. The participant was willing and able to share his or her experience of dashboard usage.

An invitation letter was sent to all school principals, assistant principals, and area superintendents who met the above criteria. The first 10 participants who met the above criteria, were willing and able to share their experience, and who were readily available for face-to-face interviews were selected. Out of the 10 participants, seven were principals, two were assistant principals, and one was an area superintendent. Six of them were males and four of them were females. The participants were not totally representative of the charter school district population.

Data collection and analysis. The data for this study were collected via face-to-face interviews or in telephone interviews with the participants. The researcher used the charter school system's administrator directory to find the contact information of school administrators. The researcher created an internet link for the interview questions for those school administrators who wanted to participate in the study. Then, the researcher sent an email to all administrators asking whether they wanted to participate in the study.

Those administrators who agreed to participate were contacted by the researcher and a time and place was set for each interview. Each interview took between 30 and 45 minutes, depending on the amount of information participants shared. All interviews were recorded on an iPhone and those recordings were later transcribed into field texts. The transcripts were reviewed and analyzed, and emerging themes from the perceptions of the administrators about the implementation of the data dashboards were identified. Quotes were used from participants' responses to answer the research questions.

Summary of major findings. The major findings of this research are presented through the research questions. Research Question One investigated what school administrators perceived as the benefits of implementing a data dashboard system for administrators, teachers, parents, and students. The study found that main benefits of using a data dashboard system are as follows:

- existing access to data,
- reviewing the school data,
- addressing community needs, and
- the advantage of accessing all data in one place

Research Question Two investigated whether school administrators felt there were ways a data dashboard system would support data-informed decision-making. The study found that the data dashboard supports the administrators for the following:

- monitoring progress,
- supporting data-informed decision-making,
- communicating with stakeholders, and
- existing access for parents.

Research Question Three asked the school administrators what they thought the data dashboard should include that was not currently being addressed. The study found that the data dashboard could be used to:

- support evaluation,
- provide cost information,
- share information with the public, and
- address all needs by including additional features.

Conclusions

This study explored the perceptions of administrators regarding the implementation of a data dashboard system in one Texas charter school system. This study affirms that understanding these perceptions about implementation will guide other charter school administrators and offer valuable insight into how to best use a data dashboard to improve the quality of education for all students in their school settings. Based on the findings from the individual interviews, specific conclusions are discussed in terms of the research questions that guided the study.

Research question one. The first research question inquired about the perceptions of benefits for implementing a data dashboard system: What do school system administrators perceive as benefits to implementing a data dashboard system for administrators, teachers, parents, and students? Based on the findings, it can be concluded that administrators now use various software to collect data for different purposes. In addition, they have to access to these different platforms through different accounts. Research findings from this study also reveal that administrators constantly need to review the data to address the needs of the students and use the information in communication with teachers, parents, and community members. This finding is consistent with the findings of several studies. Brynes (2012) mentioned that most of the executive leaders are lost within a lot of information and are seeking ways to simplify the reporting tools in an understandable and summarized format. He stated that to meet this need, many companies have developed executive-level dashboards.

Moreover, the findings in this study led to the conclusion that the data dashboard provides an advantage to its users by providing easy access to all data in one place. These

findings are also revealed in previous research which showed data dashboard tools can help school leaders make sense of critical information and get a better handle on the key measures that drive student achievement (Carr, 2008).

Research question two. The second research question focused on the perceptions of using a data dashboard system to support data-informed decision-making: What do school system administrators perceive as ways a data dashboard system supports data-informed decision-making? Research findings from this study affirm that administrators use the data dashboard to monitor the academic progress of all students, enrollment numbers, and attendance rates. Additional findings conclude that the data dashboard helps administrators make better decision-making. This supports the findings of a study conducted by West (2012). According to West (2012), the dashboard provides all the stakeholders with the critical information they need to effectively interpret and act on data in order to keep everyone on track to student, school, and district-wide goals.

Findings from this study also support that using a data dashboard helps administrators in communicating with all stakeholders. Research further reported that it was revealed that user output and dashboard applications were influential instruments in the management of performance in objectives such as decision-making justification and communication (Datnow et al., 2008). The findings from the study suggest the conclusion that the data dashboard also provides easy access to parents so that they can see what is going on at school and this helps the meaningful communication between the schools and the parents.

Research question three. The third research question asked participants to share ideas for topic expansion: What topics should the dashboard system encompass that are

currently not being addressed? The findings of this study revealed that the data dashboard could support administrators if they also have access to a teacher evaluation system through the dashboard. So, it helps administrators facilitate evaluation discussions. The findings support the study conducted by Colby et al. (2005). The researchers reported that data mining in education helps researchers examine raw data in order to understand the learning and teaching procedure (Colby et al., 2005). Following this further, the Bernhardt's (2007) and Reeves' (2005) studies demonstrated that it is very important for school leaders to facilitate professional development for their teachers in terms of data-driven decision-making processes. Moreover, they recommended that school leaders ensure professional development urges forward instructional change in the purpose of effective data-driven decision-making (Bernhardt, 2007).

Additional findings conclude that the data dashboard could provide cost information to its users so that administrators can see easily what their budgets are, what they have spent so far, and how much remains available for purchases. The findings of this study also suggest that allowing the public to have access to school data will help community members better understand the schools. This has the added benefit of providing transparency to everyone. This finding is consistent with the research of Pauwels et al. (2009). The researchers examined that when dashboards are incorporated in enterprise systems, operators can share information vertically to others.

The findings of this study also revealed that additional features, such as college-readiness data and curriculum pacing guides could be integrated into data dashboards so that administrators can get great value out of it. This supports the findings of a study conducted by Sawchuck (2010). According to Sawchuck (2010), in order to utilize

graduation provisions and reverse engineering information, government schools in New York City produced metrics to gauge whether students are on track to graduation.

Implications for Practice

The data dashboard helps school leaders gain easy access to all school related data. The wealth of information in the dashboard and the synthesis of that information on the customized dashboard significantly strengthen administrators' ability to determine the most appropriate decisions. The data dashboard also provides all stakeholders with the critical information they need to effectively interpret and act on data in order to keep them on track to student, school, and district-wide goals. The dashboard, with its easy access, also provides transparency among all stakeholders. The findings of this study provide proven practices to implement in their school systems. Suggestions to implement for practice include the following:

- Providing training for all administrators regarding the usage of a data dashboard would familiarize them with its features. Providing training for all administrators would create a culture that values data dashboard as a decision-making tool and also help in a successful implementation (Datnow et al., 2008).
- Another suggestion for implementation is to provide access to a data dashboard for teachers and train them about how to use it. This will help teachers early on to interact with data dashboard and support teachers' engagement in data-driven decision-making (Tyler, 2010).
- Arranging school nights for parents to introduce them to a data dashboard would increase the parent awareness of data dashboard features, provide

instruction for using it, and get feedback from parents for additional features to be added. Fuchs (1995) suggested that the academic progress of schools is impacted through each shareholder's input in the procedure needs, which not only involves instructional staff, but also students and their parents or guardians.

- Participating in school related conferences to share the experience of data dashboard implementation, its uses, and generate conversations about how it can best be utilized could support further use of a data dashboard system. Designing and carrying out a strategic plan to share knowledge about internal best practice would impact potential users who can most benefit from it (D'Adamo, & Kols, 2005).
- Sharing successful best practices could improve the data dashboard implementation in multiple ways. It is important to submit research results to educational journals and community newspapers to share the best practices of a data dashboard usage and describe how access to it can involve and benefit the community (D'Adamo, & Kols, 2005).

Recommendations for Future Research

This study asked only administrators to share their perceptions regarding the implementation of a data dashboard system. Other research recommendations include the following:

1. Interviewing teachers to explore their perceptions on data dashboard usage would provide more insight into how teachers can benefit from its use, and ask for their recommendations to improve it.

2. Interviewing parents to explore their perceptions on data dashboard usage would determine how they use the dashboard and whether they find it beneficial.
3. There were only 10 participants in the study. An increased number of administrator interviews would give a broader view of the uses and benefits of the dashboard and would garner more recommendations to improve it.
4. A survey of all stakeholders would generate suggestions for additional features they would like to see integrated into data dashboards.
5. A quantitative study to compare effectiveness of administrators based on their data dashboard usage.

Concluding Remarks

This study provided insight into the perceptions of administrators regarding the implementation of a data dashboard system in one Texas Charter School System. Through the process of interviewing these administrators, I learned how the data dashboard helps school administrators make data-informed decisions. As I interviewed each participant, it became obvious to me that each administrator greatly benefited from the implementation of the data dashboard. I believe that understanding these perceptions about implementation will guide other charter school administrators and offer valuable insight into how to best use a data dashboard to improve the quality of education for all students in their school settings.

References

Anderson, R., & Dexter, S. (2005). School technology leadership: An empirical investigation of prevalence and effect. *Educational Administration Quarterly*, *41*(1), 49-82. doi: 10.1177/0013161X04269517

Arnott, D., & Pervan, G. (2005). A critical analysis of decision support systems research. *Journal of Information Technology*, *20*, 67-87.

Avidov-Unger, O., & Eshet-Alkalai, Y. (2011). Teachers in a world of change: teachers' knowledge and attitude towards the implementation of innovative technologies in schools. *Interdisciplinary Journal of e-Learning and Learning Objects*, *8*(1), 291–303.

Ballou, B., Heitger, D. L., & Donnell, L. (2010). Creating effective dashboards. *Strategic Finance*, *91*(9), 27-32. Retrieved from http://www.imanet.org/PDFs/Public/SF/2010_03/03_2010_ballou.pdf

Bariff, M. L., & Lusk, E. J. (1977). Cognitive and personality tests for the design of management information systems. *Management Science, 23*(8). Retrieved May 3, 2015, from http://dx.doi.org/10.1287/mnsc.23.8.820

Benbasat, I., & Dexter, A. S. (1979). Value and events approaches to accounting: An experimental evaluation. *Accounting Review*, *54*(4) 735-749.

Bernhardt, V. L. (2007). *Translating data into information to improve teaching and learning*. Larchmont, NY: Eye on Education.

Blau, I., & Hameiri, M. (2010). Implementing technological change at school: The impact of online communication with families on teacher interactions through learning management system. *Interdisciplinary Journal of e-Learning and Learning Objects*, *6*(1), 245-257.

Blau, I., & Presser, O. (2013). e-Leadership for school principals: Increasing school effectiveness by a school data management system. *British Journal of Educational Technology, 44*(6), 1000-1011. doi:10.1111/bjet.12088

Borja, R. R. (2006). District initiative. *Education Week*, *25*(35), 24-41.

Boudett, K. P., Murnane, R. J., City, E., & Moody, L. (2005). Teaching educators how to use student assessment data to improve instruction. *Phi Delta Kappan, 86*(9), 700-706. doi: 10.1177/003172170508600915

Bremser, W. G., & Wagner, W. P. (2013). Developing dashboards for performance management. *CPA Journal, 83*(7), 62-67. Retrieved from http://www.cpajournal.com/management.htm

Brynes, J. (2012). Driving value: Solving the issue of data overload with an executive dashboard. *Healthcare Financial Management*, *66*(10), 116-118. Retrieved from http://vmlibweb.lamar.edu/

Burns, J. M. (1978). *Leadership*. New York, NY: Harper & Row.

Card, S. K., Mackinlay, J. D., & Schneiderman, B. (1999). *Readings in information visualization: Using vision to think.* San Francisco, CA: Morgan Kaufmann Publishers.

Carr, N. (2008). The data dashboard. *American School Board Journal*, *195*(12), 42-43.

Center on Education Policy. (2004). *From the capital to the classroom: Year 2 of the No Child Left Behind Act.* Washington, DC: Author.

Chamakiotis, P., & Panteli, N. (2011). e-Leadership styles for global virtual teams. In P. Yoong (Ed.)*, Leadership in the digital enterprise: Issues and challenges (pp. 143–162).* Hershey, PA: IGI Global.

Chirichello, M. (2010). The principal as educational leader: What makes the difference. In.Huber, S. G. R. (Ed.), *School leadership—International perspectives* (pp. 79–100). London: Springer.

Clark, B. H., Abela, A. V., & Ambler, T. (2006). Behind the wheel. *Marketing Management*, *25*(3), 18-23.

Cohen, F. (2003). Mining data to improve teaching. *Educational Leadership*, *60*(8), 53-56.

Colby, S., Smith, K., & Shelton, J. (2005). *Expanding the supply of high quality public schools.* San Francisco: The Bridgespan group.

Cooley, V. E., Shen, J., Miller, D., Winograd, P. N., Rainey, J. M., Yuan, W., & Ryan, L. (2006). Increasing leaders' capacity in data-based decision-making: State-level initiatives in Ohio, New Mexico, and Michigan. *Educational Horizons, 85*(1), 57–64.

Creswell, J. W. (2013). *Qualitative inquiry and research design: Choosing among five approaches* (3rd ed.). Thousand Oaks, CA: Sage.

Crommey, A. (2000). *Using student assessment data: What can we learn from schools?* Oak Brook, IL.

D'Adamo, M. & Kols, A. (2005). *A Tool for Sharing Best Practices.* The INFO Project. United States Agency for International Development (USAID), Global, GH/POP/PEC, Grant No. GPH-A-00-02-00003-00. Retrieved from: http://www.k4health.org/toolkits/km/tool-sharing-internal-best-practices

Danaher, P. J. (2002). Optimal pricing of new subscription services: Analysis of a market experiment. *Marketing Science*, *21*(2), 119-138.

Datnow, A., Park, V., & Kennedy, B. (2008). *Acting on data: How urban high schools use data to improve instruction.* Los Angeles, CA: Center on Educational Governance, University of Southern California.

Debush, G. K., Brown, R. M., & Killough, L. N. (2003). Components and relative weights in utilization of dashboard measurement systems like the Balanced Scorecard. *The British Accounting Review* , *35*(3), 214-231.

Dekimpe, M., & Hanssens, D. (1999). Sustained spending and persistent response: A new look at long-term marketing profitability. *Journal of Marketing Research*, *36*(1), 397-412.

Dexter, S. (2007). *Show me the leadership: The impact of distributed technology leadership teams' membership and practices at four laptop schools.* Paper presented at the 88th Annual Meeting of the American Educational Research Association, Chicago, IL. Retrieved from http://edtechcases.info/analysis/Dexter-AERA07-4.11.07.pdf

Doll, W. J., & Torkzadeh, G. (1998). Developing a multi-dimensional measure of system-use in organizational context. *Information & Management*, *33*(4), 171-185.

Earl, L., & Katz, S. (2006). *Leading schools in a data rich world.* Thousand Oaks, CA: Corwin Press.

Evers, M. W., & Walberg, J. H. (2002). *School accountability*. Stanford, Calif.: Hoover Institution Press, Stanford University.

Few, S. (2006). *Information dashboard design: The effective visual communication of data.* Sebastopol, CA: O'Reilly Media, Inc.

Fornell, C., Mithas, S., Morgeson, F., & Krishnan, M. S. (2006). Customer satisfaction and stock prices: High returns, low risk. *Journal of Marketing*, *70*(1), 3-14.

Fuchs, I. (1995). *Change—a way of life in schools.* Tel Aviv: Cherikover Publishers [in Hebrew].

Garrett, J. W. (2013). Technology to support data collection and management in the public schools. *Perspectives on School-Based Issues, 14*(1), 14-15. doi:10.1044/sbi14.1.10

Gleason, P., & Dynarski, M. (2002). Do we know whom to serve? Issues in using risk factors to identify dropouts. *Journal of Education for Students Placed at Risk*, *1*(1), 25-41.

Goodhue, D. L., & Thompson, R. L. (1995). Task-technology fit and individual performance. *MIS Quarterly*, *19*(2), 213-236.

Goren, P. (2012). Data, data, and more data - what's an educator to do? *American Journal of Education*, *118*(2), 233-237. doi:10.1086/663273

Gorla, N., Somers, T. M., & Wong, B. (2010). Organizational impact of system quality, information quality and service quality. *Journal of Strategic Information Systems*, *19*(3), 207-228.

Gulati, R., & Oldroyd, J. B. (2005). The Quest for Customer Focus. *Harvard Business Review*, *83*(1), 92-101.

Gurr, D. (2004). ICT, leadership in education and e-Leadership. *Discourse*, *25*(1), 113–124.

Hallinger, P. (2003). Leading educational change: Reflections on the practice of instructional and transformational leadership. *Cambridge Journal of Education, 33*(3), 329-352. doi: 10.1080/0305764032000122005

Hsieh, M. (2013). Data mining from education databases examine the factors impacting the school performance in the United States. *International Journal of Intelligent Technologies and Applied Statistics, 6*(2), 135-143. doi:10.6148/IJITAS.2013.0602.03

Huber, G. P. (1983). Cognitive style as a basis for MIS and DSS designs: Much ado about nothing? *Management Science*, *29*(5), 567-579.

Huh, Y., Keller, F. R., Redman, T., & Watkins, A. R. (1990). Data quality. *Information and Software Technology*, *29*(5), 559-565.

Ingram, D., Louis, K. S., & Schroeder, R. G. (2004). Accountability policies and teacher decision-making: Barriers to the use of data to improve practice. *Teachers College Record*, *106*(6), 1258-1287.

Jacobson, L. (2007). Districts make strides with common vision. *Education Week*, *16*(21), 28-30.

Johnson, J. H. (1999). Educators as researchers. *Schools in the Middle*, *9*(1), 38-41.

Kamakura, W., Mittal, V., Rosa, F. D., & Mazzon, J. A. (2002). Assessing the service-profit chain. *Marketing Science*, *21*(1), 294-317.

Kaplan, R. S., & Norton, D. P. (2000). Having trouble with your strategy? then map it. *Harvard Business Review*, *1*(1), 167-176.

Kaplan, R. S., & Norton, D. P. (2005). The balanced scorecard: Measures that drive performance. *Harvard Business Review*, *70*(1), 71-79.

Knapp, M. S., Swinnerton, J. A., Copland, M. A., & Monpas-Huber, J. (2006). *Data-informed leadership in education.* Seattle, WA: Center for the Study of Teaching and Policy, University of Washington.

Kowalski, T. J., Lasley, T. J., & Mahoney, J. W. (2008). *Data-driven decisions and school leadership: Best practices for school improvement.* Boston, MA: Pearson Education.

Lachat, M. A., & Smith, S. (2005). Practices that support data use in urban high schools. *Journal of Education for Students Placed at Risk*, *10*(3), 333-349.

Lange, C., Range, B., & Welsh, K. (2012). Conditions for effective data use to improve schools: Recommendation for school leaders. *International Journal of Educational Leadership Preparation*, *7*(3). Retrieved from http://files.eric.ed.gov/fulltext/EJ997478.pdf

Lapointe, P. (2005). *Marketing by the dashboard light.* New York: MarketingNPV/ Association of National Advertisers.

Lehmann, D. R., & Reibstein, D. J. (2006). *Marketing metrics and financial performance.* Cambridge, MA: Marketing Science Institute.

Little, J. D. C. (1970). Models and managers: The concept of a decision calculus. *Management Science*, *16*(8), 466-485.

Macfadyen, L., & Dawson, S. (2010). Mining LMS data to develop an "early warning system" for educators: A proof of concept. *Computers & Education*, *54*(1), 588-599. doi:10.1016/j.compedu.2009.09.008

Mandinach, E. B. (2012). A perfect time for data use: Using data-driven decision making to inform practice. *Educational Psychologist*, *47*(2), 71-85. doi:10.1080/00461520.2012.667064

MarketingNPV. (2004). Unisys overcomes 6 common dashboard mistakes. *MarketingNPV*, *1*(6), 14-16.

MarketingNPV. (2005). Timken rolls out a marketing dashboard for industrial bearing group. *MarketingNPV*, *3*(1), 3-6.

Marsh, J., Kerr, K. A., Ikemoto, G., Darilek, H., Suttorp, M., & Zimmer, R. (2005). *The role of districts in fostering instructional improvement: Lessons from three urban districts partnered with the institute for learning.* Santa Monica, CA: Rand Corporation.

Mason, S. (2002). *Turning data into knowledge: Lessons from six Milwaukee public schools.* Paper presented at the Annual Meeting of the American Educational Research Association, New Orleans, LA.

McGovern, G. J., Court, D., Quelch, J., & Crawford, B. (2004). Bringing customers into the boardroom. *Harvard Business Review*, *82*(11), 70-80.

McIntire, T. (2002). The administrator's guide to data-driven decision making. *Technology and Learning*, *21*(11), 18-33.

Mehra, A., Smith, B. R., Dixon, A. L., & Robertson, B. (2006). Distributed leadership in teams: the network of leadership perceptions and team performance. *The Leadership Quarterly*, *17*(1), 232-245.

Merriam, S. B. (2009). *Qualitative Research: A guide to design and implementation.* San Francisco, CA: Jossey-Bass.

Miller, A., & Cioffi, J. (2004). Measuring marketing effectiveness and value: The unisys marketing dashboard. *Journal of Advertising Research*, *44*(3), 237–243.

Mioduser, D., Nachmias, R., Tubin, D., & Forkosh, A. (2006). *Pedagogical innovation involving information technologies and communications.* Ramot: The Center for Science and Technology Education, School of Education, Tel Aviv University [in Hebrew].

Muntean, M., Bologa, A., Bologa, R., & Florea, A. (2011). The use of multidimensional models to increase the efficiency of management support systems. *International Journal of Mathematical Models and Methods in Applied Sciences*, *5*(8), 1334-1344.

Neely, A. D., Yaghi, B., & Youell, N. (2008). *Enterprise performance management: The global state of the art*. Retrieved from http://bit.ly/GO8Ham

Nelson, R., Todd, P., & Wixom, B. (2005). Antecedents of information and system quality: An empirical examination within the context of data warehousing. *Journal of Management Information Systems*, *21*(4), 199-235.

Paine, K. D. (2004). Using dashboard metrics to track communication. *Strategic Communication Management*, *8*(5), 30-33.

Pauwels, K., Ambler, T., Bruce, H. C., Lapointe, P., Reibstein, D., Skiera, B. (2009). Dashboards as a service: Why, what, how, and what research is needed? *Journal of Service Research*, *12*(2), 175-189.

Pauwels, K. H., Jorge, M. S., Srinivasan, S., & Hanssens, D. M. (2004). New products, sales promotions and firm value, with application to the automobile industry. *Journal of Marketing*, *68*(1), 142-156.

Pauwels, K., & Joshi, A. (2008). Counting what will count: Does your dashboard predict? *Asia Pacific Journal of Business, 2*(2), 1-35.

Peterson, J. L. (2007). Learning facts: The brave new world of data-informed instruction. *Education Next*, *1*(1), 36-42.

Pinkus, L. (2008). *Using early-warning data to improve graduation rates: Closing cracks in the education system*. Policy Brief. Washington, DC: Alliance for Excellent Education

Pyrczak, F., & Bruce, R. R. (2005). *Writing empirical research reports: A basic guide for students of the social and behavioral sciences* (5th ed.). Glendale, CA: Pyrczak Publishing.

Radas, S., & Shugan, S. M. (1998). Seasonal marketing and timing new product introductions. *Journal of Marketing Research*, *35*(3), 296-325.

Rathje, W., & Murphy, C. (2001). *Rubbish: The archeology of garbage.* Tucson: University of Arizona Press.

Reeves, D. B. (2005). *Accountability in action: A blueprint for learning organizations.* Englewood, CO: Lead + Learn Press.

Reibstein, D. J., Joshi, Y., Norton, D., & Farris, P. (2005). *Marketing dashboards: A decision support system for assessing marketing productivity.* Paper presented at the Marketing Science Conference, Atlanta, GA.

Reilly, C. (1982). Variations in decision makers' use of information sources: The impact of quality and accessibility of information. *Academy of Management Journal*, *25*(4), 756-771.

Rogers, E. M. (2003). *Diffusion of innovations* . New York: Free Press.

Rust, R. T., & Chung, T. S. (2006). Marketing models of services and relationships. *Marketing Science*, *25*(6), 560-580.

Rust, R. T., Lemon, K. N., & Zeithaml, V. A. (2004). Return on marketing: Using customer equity to focus marketing strategy. *Journal of Marketing*, *68*(4), 109-127.

Sawchuk, S. (2010). Graduation by the numbers: Putting data to work for student success. *Education Week Diplomas Count 2010, 29*(34), 19-21.

Schneider, B., Ehrhart, M. G., Mayer, D. M., Saltz, J. L., & Niles-Jolly, K. (2005). Understanding organization-customer links in service settings. *Academy of Management Journal*, *48*(6), 1017-1032.

Shen, J. (2012). *Tools for improving school principals' work.* New York, NY: Peter Lang Publishing.

Shen, J., & Cooley, V. (2008). Critical issues in using data for decision-making. *International Journal of Leadership in Education, 11*(3), 319–329. doi: 10.1080/13603120701721839

Slavin, R. E. (2002). Evidence-based education policies: Transforming educational practice and research. *Educational Researcher*, *31*(7), 15-21.

Spillane, J. P. (2012). Data in practice: Conceptualizing the data based decision-making phenomena. *American Journal of Education*, *118*(2), 113-141. doi: 10.1086/663283

Spillane, J. P., Halverson, R., & Diamond, J. B. (2001). Investigating school leadership practice: A distributed perspective. *Educational Researcher*, *1*(1), 23-28.

Srinivasan, S., & Hanssens, D. M. (2009). Marketing and firm value: Metrics, methods, findings and future directions. *Journal of Marketing Research*, *46*(3), 293-312.

Sutherland, S. (2004). Creating a culture of data use for continuous improvement: A case study of an Edison Project school. *American Journal of Evaluation*, *25*(3), 277-293.

Swanson, C. B. (2010). Pinpointing district performance. *Education Week Diplomas Count 2010. Graduation by the Numbers: Putting Data to Work for Student Success*, *29*(34), 26-27.

Tan, S., & Aloysius, O. (2011). *Distributed leadership for integration of information and communication technology (ICT) in schools.* Paper presented at Australian Society for Computers in Learning in Tertiary Education Conference, Sydney, Australia. Retrieved from http://repository.nie.edu.sg/jspui/handle/10497/4655

Tan, S. C. (2010). *School technology leadership: Lessons from empirical research.* Paper presented at Australian Society for Computers in Learning in Tertiary Education Conference, Sydney, Australia. Retrieved from http://repository.nie.edu.sg/jspui/handle/10497/4622

The Children's Online Privacy Protection Act of 1998 (15 U.S.C. § 6501-6506) (Pub.L. 105–277, 112 stat. 2581-728, enacted October 21, 1998).

The Information Edge. (2006). Using data to accelerate achievement. *Education Week 25*(35).

Tim, A. (2003). *Marketing and the bottom line.* London: Financial Times Prentice Hall.

Tomic, B., & Milic, T. (2013). Automated interpretation of key performance indicator values and its application in education. *Knowledge-Based Systems, 37*(2013), 250-260. doi: 10.1016/j.knosys.2012.08.006

Tyler, John H. (2010). *Lessons from Cincinnati's implementation of the data dashboard*. Unpublished manuscript, Department of Education, Brown University, Providence, RI.

Vessey, I., & Galetta, D. (1991). Cognitive fit: An empirical study of information acquisition. *Information Systems Research*, *2*(1), 63-84.

Wayman, J. C. (2005). Involving teachers in data-driven decision making: Using computer data systems to support teacher inquiry and reflection. *Journal of Education for Students Placed at Risk*, *10*(3), 295-308.

Wayman, J. C. (2007). *Student data systems for school improvement: The state of the field.* Symposium conducted at the meeting of the TCEA Educational Technology Research, Lancaster, PA.

Wayman, J. C., Stringfield, S., & Yakimowski, M. (2004). *Software enabling school improvement through analysis of student data.* Baltimore, MA: The Center for Research on the Education of Students Placed At Risk.

West, D. M. (2012). *Big data for education: Data mining, data analytics, and web dashboards.* Retrieved from http://www.insidepolitics.org/brookingsreports/education%20big%20data.pdf.

Whiting, R. (2006). *Case study: Using data mining to analyze student behavior*. Retrieved May 3, 2015, from http://www.schoolcio.com/showarticle/902

Wierenga, B., & Bruggen, G. H. (2000). *Marketing management support systems: Principles, tools and implementation.* Boston: Kluwer Academic.

Wind, Y. (2005). Marketing as an engine of business growth: A cross-functional perspective. *Journal of Business Research*, *58*(1), 863-873.

Wirtz, J., Mattila, A. S., & Lwin, M. O. (2007). How effective are loyalty reward programs in driving share of wallet? *Journal of Service Research*, *9*(4), 327-334.

Wyner, G. (2008). Marketing effectiveness: Practical approaches to measurement and management. *Marketing Knowledge Series*, *1*(1), 41-76.

Yigitbasioglu, O. M., & Velcu, O. (2012). A review of dashboards in performance management: Implications for design and research. *International Journal of Accounting Information Systems*, *13*(1), 41-59.

Ying, L., Lijun, X., & Wei, S. (2009). *Designing supply chain KPIs for upper-level management*. Paper presented at ITTA International Conference, Zhangjiajie, China. Retrieved from http://bit.ly/GY7FSA

Zeithaml, V. A., Bolton, R. N., Deighton, J., Keiningham, T.L., Lemon, K. N., & Peterson, J. A. (2006). Forward-looking focus: Can firms have adaptive foresight. *Journal of Service Research*, *9*(2), 168-183. doi: 10.1177/1094670506293731

List of Appendices

Appendix A Guided Protocol

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Appendix A

Guided Protocol

The purpose of this qualitative study is to explore the perceptions of administrators about the implementation of a data dashboard system in one Texas Charter School System. School leaders always need tools such as a data dashboard to help them identify teachers' performance and why students are not mastering specific concepts (Blau & Presser, 2013). They need a system that provides real-time information to users, including students, through customized user-friendly dashboards. Creating a data dashboard will help school leaders in this purpose (Ballou et al., 2010). When this information is used to guide the decisions that are in the best interest of students and families we serve, it becomes knowledge (West, 2012).

The following research questions will be used in this study:

1. What do school system administrators perceive as benefits to implementing a data dashboard system; in terms of administrators, teachers, parents, and students?
2. What do school system administrators perceive as ways a data dashboard system supports data-informed decision-making?
3. What topics should the dashboard system encompass that are currently not being addressed?

Purpose Statement: The purpose of this qualitative study is to explore the perceptions of administrators about the implementation of a data dashboard system in one Texas Charter School System.

Background and Get Acquainted Questions:

A. Tell me about yourself. What is your name? What is your position at the school?

B. How long have you been in Education? Describe yourself and your interest as an educator.

C. How many years of experience as school administrator?

Research Questions:

1. What do school system administrators perceive as benefits to implementing a data dashboard system for administrators, teachers, parents, and students?
    A. What are the internal needs for creating a data dashboard? How would having access to a data dashboard for your campus impact your school?
    B. What are the external needs for creating a data dashboard? How would having access to information in a data dashboard for your campus impact your community/parents/stakeholders?
    C. When you have access to data, how do you interpret it? How do you use data on your campus?
2. What do school system administrators perceive as ways that a data dashboard system supports data-informed decision-making?
    A. How do you use data in conversations with parents and staff?
    B. How do you use data to identify persistent problems and monitor progress of school over time?

C. Please talk about the school data to which your parents have ready to access. What other data would you like for parents to be able to access?

3. What topics should the dashboard system encompass that are currently not being addressed?

   A. How could a data dashboard for your campus be used to support evaluation?

   B. How could a data dashboard for your campus be used to provide cost information?

   C. How could a data dashboard for your campus be used to share information with the public?

Appendix B

Letter of Invitation to Participate in Study

Date : 1/6/2016

Name : TBD

Address: TBD

Dear Participant:

As a doctoral candidate at Lamar University, I am conducting a study for my dissertation to understand and explore the perceptions of administrators about the implementation of a data dashboard. The purpose of this qualitative phenomenological narrative study is to explore the perceptions of administrators about the implementation of a data dashboard system in one Texas Charter School System. I will be examining the benefits of implementing a data dashboard system, in what ways data dashboard system supports data-informed decision making, and what topics the dashboard system should encompass that are currently not being addressed. The results of the study should be of interest and value to the leadership of school administrators, teachers, parents and students who are using data dashboard system as a tool to share key performance indicators and any other relevant information to all stakeholders in real time and easy format.

Approximately 10 people will be invited to participate in this study. Criteria for participation include the following:

1. The participant is a charter school administrator serving as a principal, assistant principal, or area superintendent.
2. Each administrator had two or more years of experience in the district and in their current position.
3. The participant was willing and able to share his or her experience on dashboard usage.

During the first week in January 2016, you will receive an e-mail whether you would like to participate in the study. The email will include a link to the interview

questions and guided protocol. The researcher will contact those who agree to participate in the study for a time and place for the interview. The interview will require approximately 45 minutes to an hour to complete all the protocol questions. Once the interview recordings were collected, the recordings were transcribed. Then, they were coded and the researcher analyzed the responses to discover common patterns within the perceptions of the administrators about the implementation of the data dashboards. Your responses will be completely anonymous, and no individual names or institutions will be recorded during the course of this study. Your name will not be associated with any research findings. The data will be stored electronically in a secure place.

This study has been approved by both the Charter School District's Institutional Review Board and the Lamar University Institutional Review Board. You are free to decide not to participate in this study or to withdraw at any time without adversely affecting your relationship with the researcher, the charter school district or Lamar University. There is no compensation for participating, however, on completion of this study, I will share a summary of the findings with you. Your input is extremely valuable for the preparation of future educational leaders in Texas. If you have any questions regarding the study, please contact the Principal Researcher or the Dissertation Chair person.

Sincerely,

Principal Researcher: Erdal Caglar
Contact information (e-mail): erdalcaglar@gmail.com
Dissertation Chair: Dr. Diane Mason
Contact Information: diane.mason@lamar.edu

Appendix C

NIH Certificate

Certificate of Completion

The National Institutes of Health (NIH) Office of Extramural Research certifies that **Erdal Caglar** successfully completed the NIH Web-based training course "Protecting Human Research Participants".

Date of completion: 09/11/2014

Certification Number: 1552200

Appendix D

IRB Notice of Approval – Lamar University

Memorandum

Office of Research & Sponsored Programs
Institutional Review Board
Lamar University

Date: January 6, 2016
To: Erdal Caglar
From: Office of Research & Sponsored Programs Administration
Re: Request for approval by IRB

Your project, "Administrators' perceptions of the implementation of a data dashboard system in one Texas charter school system" was reviewed and approved. It qualifies for exemption because the research employs standard methods and procedures for testing and educational purposes, involves minimal risks to subjects, and makes participation strictly voluntary. This approval is for a period ending one year from the date of this memorandum. Please make timely submission of renewal or prompt notification of project termination. Your IRB # is 73416112.

Remember to obtain approval from the Institutional Review Board before instituting any changes in the project. The Board wishes you every success in your research endeavor.

Biographical Note

Erdal Caglar graduated from the Bosphorus University in Istanbul and received his Bachelor of Science in Teaching Mathematics in 1995. He taught mathematics for several years. He attended University of Texas at Tyler and received his Master's Degree in Educational Leadership in 2012. He was accepted into the Doctoral Cohort VI at Lamar University where he earned a Doctorate of Educational Leadership in 2016. Currently, he continues to serve as a Chief Information Officer at a Charter School System in Texas.

Permanent Address: 4039 Mission Valley Dr. Missouri City, Texas 77459

Style manual delegation: *Publication Manual of the American Psychological Association, Sixth Edition*

Typist: Erdal Caglar